forests

MDPI

Review

The Impact of Forest Fires on Ecosystem Services: The Case of Greece

Anastasia Misseyanni [1,2], Anastasia Christopoulou [2,3], Ioannis Kougkoulos [1], Emmanuel Vassilakis [4] and Margarita Arianoutsou [2,*]

1 Department of Science and Mathematics, Deree—The American College of Greece, 15342 Athens, Greece; amisseyanni@acg.edu (A.M.); ikougkoulos@acg.edu (I.K.)
2 Department of Ecology and Systematics, Faculty of Biology, National and Kapodistrian University of Athens, 15784 Athens, Greece; anchristo@biol.uoa.gr or achristop@aegean.gr
3 Biodiversity Conservation Laboratory, Department of Environment, University of the Aegean, 81100 Mytilene, Greece
4 Department of Geography and Climatology, Faculty of Geology, National and Kapodistrian University of Athens, 15784 Athens, Greece; evasilak@geol.uoa.gr
* Correspondence: marianou@biol.uoa.gr

Abstract: (1) Background: The present study examines the effects of fire on the ecosystem services of forest ecosystems in Greece. Being a Mediterranean country, Greece has been affected by fires of increasing intensity and frequency in recent years; (2) Methods: Information was extracted from 56 articles published in the period January 1997–March 2024 that were selected after an extensive literature review; (3) Results: An increasing trend in the number of published articles over time was observed. Studies on regulating and maintenance services prevailed. The majority of studies reported on thermo-Mediterranean ecosystems, with *Pinus halepensis* Mill forests being the most common ecosystems affected by fires. The effects of fire were primarily negative on provisioning and cultural services, as well as on the control of erosion rates, regulation of the hydrologic cycle, atmospheric composition, and climate regulation. Most effects on plant diversity were found to be positive, while positive and neutral effects were also recorded for pollination. The most pronounced negative or positive effects were noted for the first two years after the fire. The spatial mapping of the results showed that the areas most affected by the fires in Greece are Eastern Attica, Euboea, Western Attica, and most regional units of the Peloponnese; (4) Conclusions: In the era of climate change and changing fire regimes in the Mediterranean, there is a need to further research the impact of fire on ecosystem services, as this will help in the better protection and management of the most vulnerable forest ecosystems.

Keywords: wildfire; Mediterranean forest ecosystems; ecosystem services; regional units; Greece

check for updates

Academic Editor: Panteleimon Xofis

Received: 7 February 2025
Revised: 5 March 2025
Accepted: 12 March 2025
Published: 17 March 2025

Citation: Misseyanni, A.; Christopoulou, A.; Kougkoulos, I.; Vassilakis, E.; Arianoutsou, M. The Impact of Forest Fires on Ecosystem Services: The Case of Greece. *Forests* **2025**, *16*, 533. https://doi.org/10.3390/f16030533

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Forests cover about one-third of the Earth's surface and provide valuable ecosystem services. They mitigate the effects of climate change and act as important carbon sinks [1], removing greenhouse gases from the atmosphere [2,3]. They support life on Earth, hosting the vast majority of species that live on land, they protect the soil from erosion, they regulate the hydrologic cycle, they offer important provisioning services as sources of biomass (food, fuel, and pharmaceuticals) and water, and they also provide cultural services, which are associated with their recreational, esthetic, and spiritual value [4]. According to the Food and Agriculture Organization, forests contribute to all 17 goals of sustainable

development [5], being at the heart of Sustainable Development Goal 15 [6], which aims to protect, restore, and promote sustainable use of terrestrial ecosystems and to sustainably manage forests, among other.

Contemporary environmental challenges pose a threat to forest ecosystems and the services they provide. Among others, a forest's capacity to absorb carbon seems to be negatively affected by climate change [7–9] and the increasing forest fire emissions linked to it [10,11]. EU countries will need to increase their forest carbon sinks to reach their climate neutrality goal by 2050 [12]. Climate change and changes in land use lead to an increase in phenomena such as extreme rainfall and fire, that negatively impact the ability of forests to control erosion rates and regulate the hydrologic cycle [13,14]. The risk of increasing erosion rates, floods, and landslides is particularly high in countries with complex terrain, and forested land on slopes [14,15].

Mediterranean forests include coniferous forests of thermophilous species, found at low elevations, with sclerophyllous shrubby species in their understory, deciduous forests, as well as coniferous forests at higher elevations. Among these, Mediterranean pine forests (including thermophilous species such as *Pinus halepensis* Mill., *Pinus pinaster* Ait., and *Pinus brutia* Ten. and montane species such as *Pinus nigra* J.F. Arnold) are important sources of wood. In Spain, the main destination of cut wood is timber, while in Southern Mediterranean the most important forest product is firewood, which has potential to cover renewable energy needs [16]. However, it should be noted that firewood's renewable and "carbon neutral" nature depends on how fast forests regrow [17,18] and that biomass burning also emits particulates affecting human health [19] and brown carbon which contributes to radiative forcing [20]. Mediterranean pine forests further provide non-wood forest products such as resin (from *Pinus pinaster* and *P. halepensis*), pine nuts (from *Pinus pinea* L.), mushrooms, and pine honey (from *Pinus halepensis* and *P. brutia*) [21–23]. They additionally provide important regulating services such as carbon sequestration and protection of soil from erosion [24], being used in reforestation and afforestation projects to restore and stabilize degraded land; they harbor biodiversity of high conservation value [23]. Hunting, recreation, educational, and scientific activities are among the most important cultural services offered by pine forests [23]. The number of research studies on forest ecosystem services at global level has rapidly increased over the past two decades [25,26].

The ecosystems of the Mediterranean region are prone to fire, which is normally induced by the hot and dry conditions of the prolonged summer period. Fire is a disturbance that has acted as an important evolutionary pressure and has shaped plant diversity in Mediterranean ecosystems, as many plant species have developed fire-adaptive traits, such as serotinous cones, thick bark, the ability to form resistant seed banks that germinate after the fire, or the ability to resprout after the fire [27–31]. In recent decades, the changing fire regime in the Mediterranean region [32,33] has been associated with the changing climate, but also with changes in land use and land cover, such as the abandonment of rural land, which gradually becomes shrubland and forest, the accumulation of dry biomass due to lack of forest management, and the expansion of the wildland-urban interface that exposes forests close to urban centers to the impact of human activities [23,34].

Fire is a natural disturbance that has an impact on ecological processes. It is of critical importance to understand its impact on forest ecosystem services, so that appropriate measures can be taken for their management and conservation. Studies on the effects of fire on specific ecosystem functions or services have been conducted for different ecosystem types such as savannas [35], mountainous coniferous forests in North America [36–38], and Mediterranean-type ecosystems [31,39–42]. Fire negatively impacts many regulating services of forests. Many studies confirm the negative effects of fire on the control of soil erosion rates [43–46] and the regulation of the hydrologic cycle [36,46], as the soil of a burned

area may lose its ability to regulate mass flows and may lead to extreme phenomena. Fire also affects the physicochemical and biological properties of the soil, causing changes in its texture, pH, nutrients, organic matter, and microbial and faunal community [43,47–50]. The effects of fire on pollination, a regulating service of critical importance for the maintenance of the life cycle, have been discussed on a global scale [43,51]. The open spaces created by the fire promote the establishment of herbaceous species, creating in this way an environment that is favorable for pollinators [43,52,53]. Biodiversity, a fundamental ecosystem component contributing to the maintenance of the life cycle is significantly affected by fire. Studies in different types of Mediterranean coniferous forests, of low and high elevation, have shown an increase in plant diversity in the first years after the fire, with species that either resprout or regenerate via seed germination [40,41]. Ecological succession in Mediterranean forests follows the initial floristic composition model and the majority of species found in the final community appear immediately after the fire, while the dynamics of succession are determined by changes in their abundance and cover [54,55]. Wildfires affect the ability of forests to regulate the climate, a critical ecosystem service, as they release large amounts of greenhouse gases and reduce tree biomass, thus converting the forest from a carbon sink to a carbon source and aggravating climate change impacts, as confirmed by a number of studies [10,11,43,56]. The ability of forests to regulate atmospheric quality is also impaired through the emissions of atmospheric pollutants during a wildfire [43,56]. Wildfires are also expected to affect cultural services provided by forests, as they may change a landscape for decades. They are usually considered the worst natural disaster that affects tourist arrivals [57]. Negative impacts of fires on ecotourism have been reported [45], while positive impacts on cultural services have also been found [38].

Fire as a natural phenomenon has shaped ecosystems and biodiversity over the course of time. In an opinion article by Pausas and Keeley [58], fire is presented as an ecosystem service, as, from an evolutionary and socioeconomic perspective, it provides a number of benefits, including the control of pests, increasing the provision of water, accelerating species replacement, and positively impacting pollination, while frequent, controlled fires prevent against larger crown fires. The authors note, though, that many current anthropogenic fire regimes may lead to catastrophic fires [58].

The Mediterranean basin is not only a biodiversity hot spot but also a climate change hot spot, warming faster than the global average [59]. This will have inevitable consequences for the natural capital and the ecosystem services of Greek forests [60]. One such consequence is the change in the fire regime, making the region a hot spot for forest fires. Greece has been particularly affected by fires in the last two decades, with the biggest losses in burnt areas having been recorded in the years 2007, 2021, and 2023, according to the European Forest Fire Information System [32]. The forests of Greece include areas of high conservation value and correspond to 30 different habitat types of Directive 92/43/EEC [61]. Climate change is expected to increase the fire risk in the Mediterranean [62] and affect forest ecosystems [63,64]. While thermo-Mediterranean tree species show resilience to fire, montane tree species lack fire-adaptive traits and are thus most threatened by changes in the fire regime [65–67]. For the above reasons, the study of the impact of fire on forest ecosystem services is becoming particularly relevant today.

The purpose of this paper is to provide an overview of studies on the impact of fire on forest ecosystem services in Greece, to examine which ecosystem services have been found to be affected more and in what ways, and to examine how factors such as the time span since the last fire and the type of ecosystem might affect these impacts. As no prescribed fires are used for forest management in Greece, all the studies examined herein refer to wildfires; in nearly all cases, these were stand-replacing, crown fires, in which the above-ground plant biomass was completely consumed.

In times of climate change and changing fire regimes, it becomes particularly important to study the impact of forest fires on ecosystems and human societies; emphasis on the ecosystem services aspect of forests may help raise better awareness among forest managers, policy makers, and the public and may perhaps lead to better management practices. The findings of our study can support decisions concerning the protection and management of the forest ecosystems of Greece, especially those with the highest vulnerability and value.

2. Materials and Methods

2.1. Data Collection

The employed methodology included a literature review on the impact of forest fires on ecosystem services, with focus on Greece. Information was taken from peer-reviewed studies (journal articles, book chapters, and articles from conference proceedings) that were published in the last three decades (period 1 January 1997–10 March 2024) in English, and the search was performed through Web of Science, Scopus, and Google Scholar databases. We searched for relevant publications by title, abstract, and keywords. As we did not expect to find many studies that combine the words "forest fires" and "ecosystem services" referring to Greece, we decided to infer our results from studies on the impact of forest fires on ecosystem structure, functions, and processes. The keywords used in the search were "wildfire" or "fire" or "burn", "Forest" or "woodland" or "tree", "ecosystem service *" or "ecologic* process*" or "ecologic * function *" or "provision *" or "regulat *" or "cultural", and "Greece" or "Greek". After merging and deduplication, 292 publications were collected on 10 March 2024. In the first screening stage, where abstracts and titles were examined, the publications that did not refer to a specific ecosystem service category were excluded. In the second screening stage, at which 108 full-text articles were examined and data were evaluated, we excluded publications that included data that (a) did not correspond well to a specific ecosystem service indicator according to the Common International Classification of Ecosystem Services (CICES vs. 5.1) and/or (b) did not allow clear assessment of the impact of fire (e.g., the difference in the ecosystem service indicator between burned and unburned areas). Additional searches, using the same exclusion criteria, were conducted using other methods. After screening, 56 publications from all sources were included for analysis (54 from databases and 2 from other sources). The search procedure is shown in the PRISMA flow diagram of Figure S1 [68].

Given that the effect of fire on ecosystems may vary based on time after the fire incident, effects were separated into short-term (1–2 years after the fire), medium-term (3–20 years after the fire) and long-term (>20 years after the fire). The first two years after a fire are critical for the emergence and the survival of the pine seedlings, while the limit of 20 years was selected in order to include the entire time span that is usually required for a thermo-Mediterranean pine forest to complete its transition to the reproductive phase [69]. For the classification of ecosystem services, we followed the most recent version of the Common International Classification of Ecosystem Services (CICES vs. 5.1) and we distinguished them into three main categories: provisioning, regulating and maintenance, and cultural services, with regulating and maintenance services being further divided as follows: (a) control of erosion rates, (b) regulation of the hydrologic cycle, (c) biodiversity-lifecycle maintenance, (d) pollination, (e) regulation of soil quality, and (f) atmospheric composition and climate regulation. This led to a total of eight (8) ecosystem services categories. Forest ecosystem types were distinguished into two major categories encountered in Greece: thermo-Mediterranean, with main representatives Mediterranean pine species adapted to hot and dry conditions, such as *Pinus halepensis* and *Pinus brutia*, and montane Mediterranean (oro-Mediterranean) species, which are less resilient to fires, with the main representative species being *Abies cephalonica* Loudon and *Pinus nigra* [28,70].

Studies which examined areas with a combination of the two types of ecosystems or with a broad spectrum of ecosystems and land uses were assigned to different ecosystem categories, which led to a total of four categories of ecosystem types. We characterized the effects of fire on an ecosystem service as positive when an increase in the service was observed, negative when there was a decrease in the service, and neutral when no statistically significant effects on the ecosystem service were observed, always in relation to the pre-fire condition.

2.2. Data Analysis

Since prescribed fires are not used in Greece for management purposes, all 56 publications analyzed examined only wildfire impacts. Information from the selected articles was analyzed and a database was produced which contained the following fields: (i) area of study, (ii) ecosystem type, (iii) year of fire, (iv) time span since last fire, (v) ecosystem service(s) studied, (vi) parameter(s) measured/assessed, (vii) method of assessment, and (viii) impacts of the fire (positive, negative, neutral). Information on fire frequency, fire severity, and treatments after the fire was also recorded. From the total of 56 articles that were examined, five studies described the impact of fire on two different ecosystem services and one studied the impact on three different services, which led to a total of 63 studies of fire impacts. For studies that examined more than one ecosystem service, different timeframes since the last fire, different locations, different fire frequencies, or different treatments after the fire, more entries were created in the database (a separate entry for each variation in the above factors), leading to a total of 175 entries.

2.3. Geospatial Data Analysis

The geospatial representation took place at the level of the regional unit. Greece is divided into 13 regions (main administrative units of the country) which are further divided to a total of 74 regional units. Regional units were considered an appropriate unit of spatial reference because in some studies there was not enough spatial information available to allow analysis at a smaller spatial scale.

The software ArcGIS Pro 3.4.2 was used for the geospatial representation of the results. For creating the maps, we used the initial database and enriched it with spatial information. More specifically, the regional units of all areas of the study were identified, and the new database that was created for the GIS mapping included the following fields for each study: (i) the regional unit (s) examined (articles that referred to the entire country were not included), (ii) the ecosystem service (s) studied, (iii) the time span since the last fire incident (3 categories, as described in Section 2.2), and (iv) the impact of the fire on each ecosystem service, with one entry per regional unit. A separate category was created for conflicting impacts in the same regional unit, which led to a total of four categories of impacts.

3. Results

3.1. Characteristics of the Studies Examined

Both the number of studies and the number of entries show variation in the examined period (1997–2024), with maximum number of studies recorded in years that followed big fires, such as 2010 (after the fires of 2007) and 2022 (after the fires of 2021) (Figure 1).

The largest number of studies (86%) and entries (94%) refer to regulating services, while much lower numbers refer to provisioning and cultural services (Table S1 in Supplementary Materials). As far as regulating services are concerned, the majority of studies examined the impact of fire on controlling erosion rates (18 studies, 47 entries), followed by their impact on biodiversity (15 studies, 65 entries), while a smaller number of studies and entries examined the impacts of fire on atmospheric composition and climate

regulation, regulation of the hydrologic cycle, regulation of soil quality and pollination services (Figure 2). A variety of methods, both biophysical and socioeconomic, were used for assessing ecosystem services.

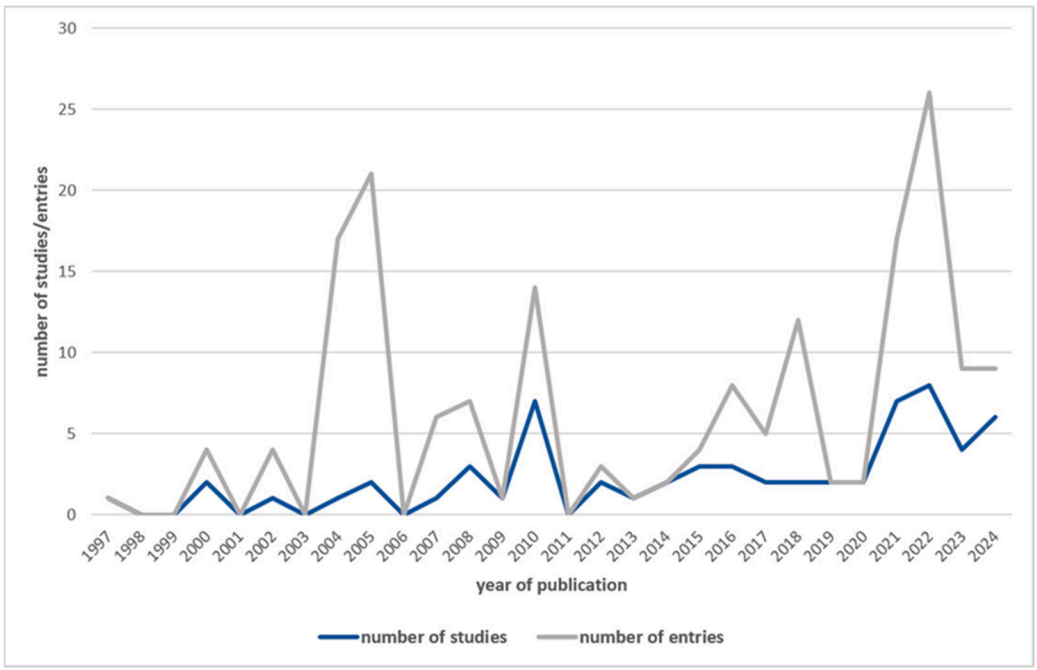


Figure 1. The number of studies and entries of fire impacts per year, period January 1997–March 2024.

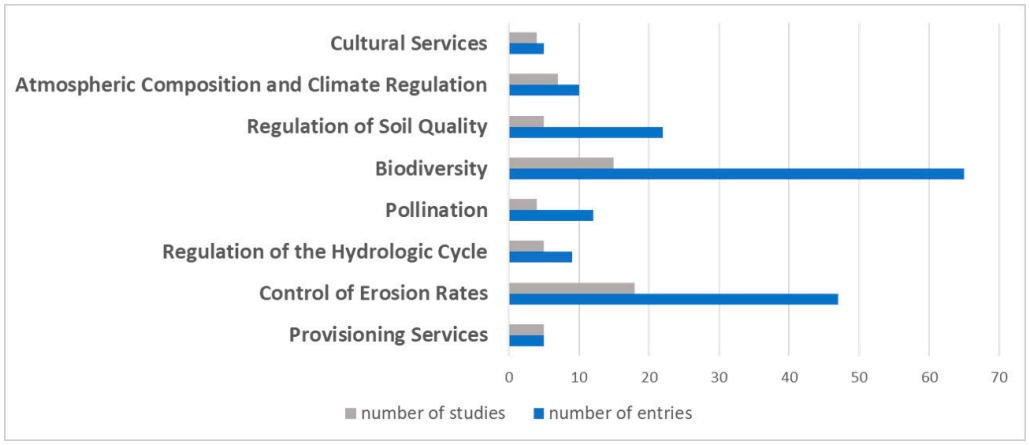


Figure 2. The number of studies and entries per ecosystem service.

The vast majority of studies and entries refer to thermo-Mediterranean forests (79% of all entries), while only 7% of entries refer to oro-Mediterranean forests and 11% to combinations of these or to a broader spectrum of ecosystems and land uses (Figure 3a). The number of entries for each ecosystem type and ecosystem service category is shown in Figure S2 in Supplementary Materials. The majority of studies and entries (65% of all entries) focused on the short-term effects of fire on ecosystem services, while about a quarter of the entries (27%) referred to the medium-term effects, and only a few studies referred to the long-term effects (8% of all entries) (Figure 3b). A more detailed representation of the number of entries on short-term, medium-term, and long-term effects per ecosystem service is shown in Figure S3.

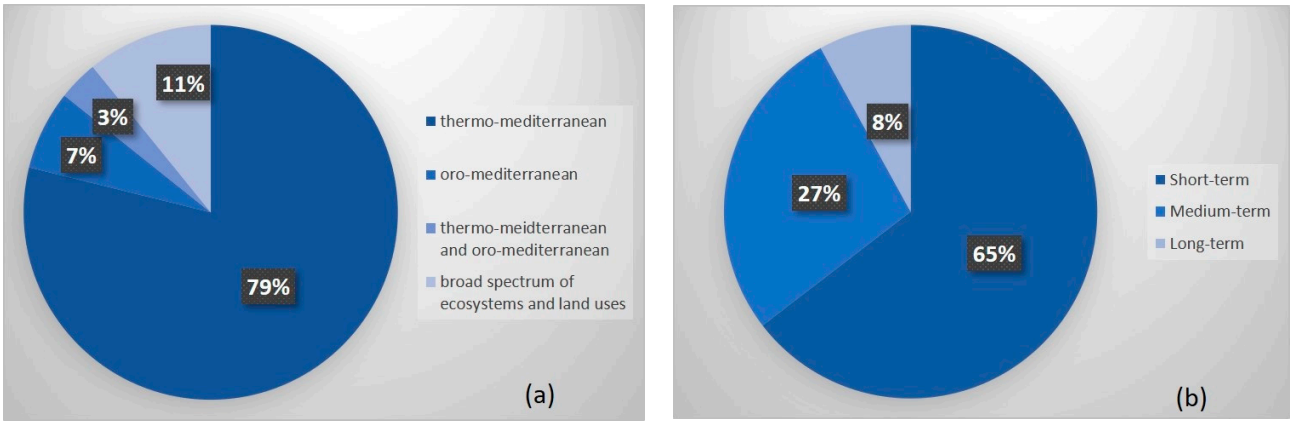


Figure 3. (**a**) Percentage of entries per ecosystem type; and (**b**) percentage of entries vs. time span since last fire.

As far as the spatial distribution of studies is concerned, most studies related to regulating and maintenance services were conducted in the regional unit of East Attica (17 studies), followed by the regional units of Euboea (9), Laconia (6), West Attica (6), Achaea, and Messenia (5 studies each), Arcadia, Elis, and Corinthia (4 studies each), Chios (3), Chalkidiki and Rhodes (2 studies each), Thasos, Thessaly, Phthiotis, Zakynthos, Central Athens, South Athens, and Lesbos (one study each) (Figure 4). The studies in East Attica, Euboea, and the regional units of the Peloponnese refer mainly to the 2007 fires in Attica (Mt. Parnitha), the Peloponnese, and Euboea, the 2021 fires in Attica and Euboea, as well as the multiple fires in Mt. Penteli (Attica) during the period of 1982–2009. The regional units that were mostly studied for provisioning services were those of Euboea (2021 fire), while the regional units of Peloponnese that were hit by the fires of 2007 (Corinthia, Achaea, Arcadia, Messenia, and Laconia) were studied for the effect of fire both on provisioning and cultural services (Figure 4).

3.2. Impacts of Forest Fires on Ecosystem Services in Greece

3.2.1. Analysis of Fire Effects

The impacts of fire on forest ecosystem services are summarized in Figure 5. For ecosystem services such as the control of erosion rates, regulation of the hydrologic cycle, atmospheric composition, and climate regulation, the negative impacts prevail, while for biodiversity and pollination—two ecosystem services associated with life cycle maintenance—the number of positive impacts recorded is higher; the impacts on the regulation of soil quality are mostly neutral (Figure 5).

More specifically, all entries for provisioning services referred to short-term effects, showed mostly negative effects (80% of entries), and examined parameters related to food and resin production as well as animal husbandry (Figure S4, Table S2). For the control of erosion rates, most entries referred to short-term effects, and these were all negative, while in the fewer studies that examined medium-term and long-term effects, negative and neutral effects were recorded (Figure S5, Table S2). As for the effects on the regulation of the hydrologic cycle (runoff and flood risk), short-term and medium-term effects were examined and they were all negative (Figure S6, Table S2). Studies on the impact of fire on pollination services examined the abundance and diversity of pollinating insects and showed positive short-term effects, positive or neutral medium-term effects, and neutral long-term effects (Figure S7, Table S2). The impact of fire on biodiversity was characterized by the largest number of entries (Figure 2). Plant diversity was mostly examined (54 out of 65 total entries on biodiversity), with few entries on soil arthropod diversity (7 entries out of 65 total) and bird diversity (4 entries out of 65 total) (Figure S8, Table S2). The

ecosystem type mostly examined in these studies is thermo-Mediterranean pine forests with *Pinus halepensis* as the main tree species (85% of all entries) (Figure S9). The studies on the impact of fire on the regulation of soil quality examined short-term and medium-term effects on parameters such as soil pH, organic matter, decomposition rate, and soil fertility. Neutral effects prevailed, but some negative (5/18 entries) and a few positive (2/18 entries) short-term effects were also recorded (Figure S10, Table S2). The entries on fire impacts on atmospheric composition and climate regulation showed negative short-term and medium-term effects and neutral long-term effects (Figure S11). The parameters examined were atmospheric composition and quality, greenhouse gas emissions, and Gross Primary Productivity (GPP) (Table S2). As far as the effects of fire on cultural services are concerned, all entries showed negative short-term and medium-term effects (Figure S12). Cultural services examined were cultural heritage, tourism, esthetic value, and hunting demand (Table S2).

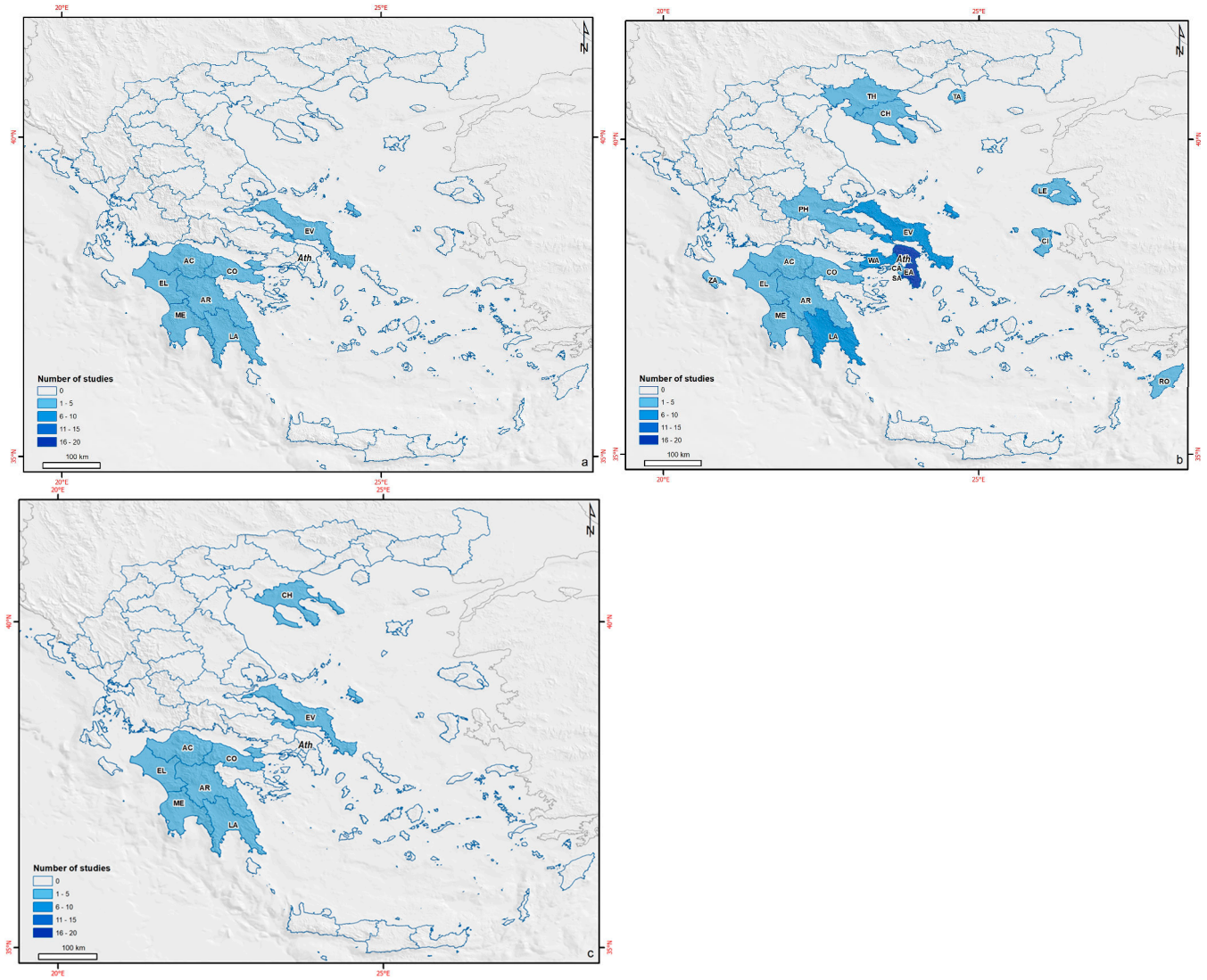


Figure 4. Geospatial distribution of studies on (**a**) provisioning, (**b**) regulating and maintenance, and (**c**) cultural services, compiled by Greece's regional units. Abbreviations for regional units are as follows: AC, Achaea; AR, Arcadia; CA, Central Athens; CH, Chalkidiki; CI, Chios; CO, Corinthia; EA, East Attica; EL, Elis; EV, Euboea; LA, Laconia; LE, Lesbos; ME, Messenia; PH, Phthiotis; RO, Rhodes; SA, South Athens; TA, Thasos; TH, Thessaloniki; WA, West Attica; ZA, Zakynthos. Ath stands for the city of Athens.

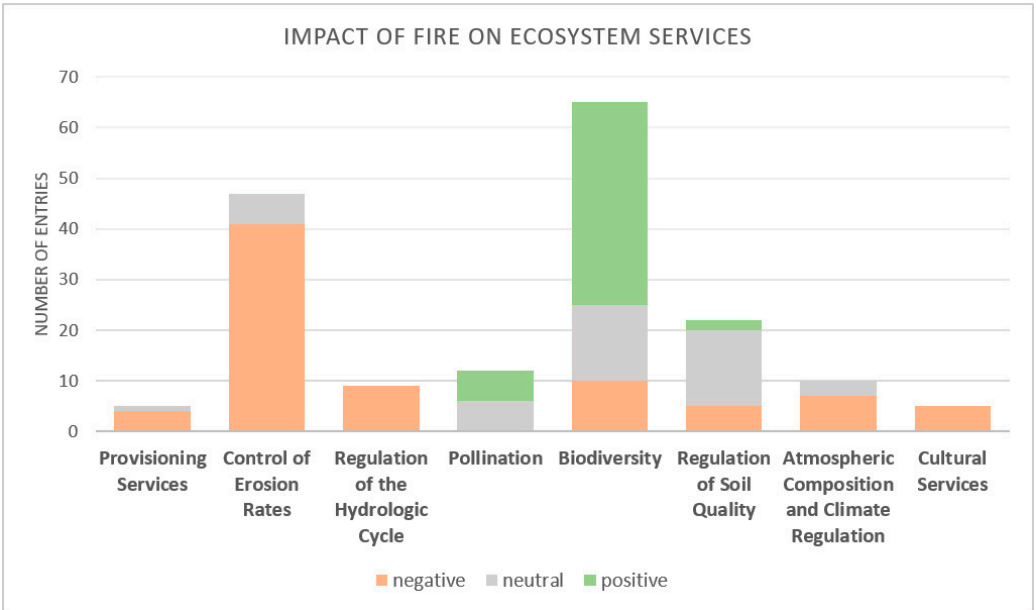


Figure 5. The number of entries showing positive, neutral, and negative effects of fire on the eight selected ecosystem services.

3.2.2. Geospatial Analysis

The results of the geospatial analysis are summarized in Figure 6. Long-term impacts are not represented, due to the low number of studies. Concerning the impacts of fire on the control of erosion rates, the regional units that suffered negative short-term effects were those of Euboea, East Attica, Corinthia, Achaea, Elis, Messenia, and Laconia (Figure 6b). Some of the medium-term effects were found to be neutral (regional units of East Attica, Phthiotis, and Thessaloniki) (Figure 6c). In a similar way, all effects on the regulation of the hydrologic cycle were found to be negative in the regional units of East Attica, Arcadia, and Laconia (Figure 6d,e). All effects on atmospheric composition and climate regulation were also negative, with short-term studies in Euboea, East Attica, West Attica, and all regional units of Peloponnese except Argolida (Figure 6l,m). Negative effects on provisioning and cultural services were recorded in the regional units of Euboea and many regional units of Peloponnese (Figure 6a,n,o).

For pollination and biodiversity, two services tightly associated with the maintenance of the life cycle, the picture is different. Biodiversity has been studied extensively in the regional units of East Attica, Euboea, West Attica, Achaea, Arcadia, Messenia, Laconia, Chalkidiki, and Chios, with positive short-term effects (as far as plant diversity is concerned), which in some regional units become neutral in the medium term (Figure 6h,i). Positive effects of fire on pollination services (both short-term and medium-term) have been recorded in the regional unit of Chios, while the medium-term effects of fire in the regional unit of Rhodes are neutral (Figure 6f,g). As far as the regulation of soil quality is concerned, the impacts of fire were neutral overall in East Attica, while the studies in Euboea and Thessaloniki showed different types of effects, depending on study (positive, negative and neutral) (Figure 6j,k).

In conclusion, and after taking into account all studies, it seems that the regional units of East Attica, Euboea, and West Attica as well as those of Peloponnese which were hit by the 2007 fire (Corinthia, Achaea, Elis, Messenia, Laconia, Arcadia) are the most negatively impacted areas by fires that took place before 2023.

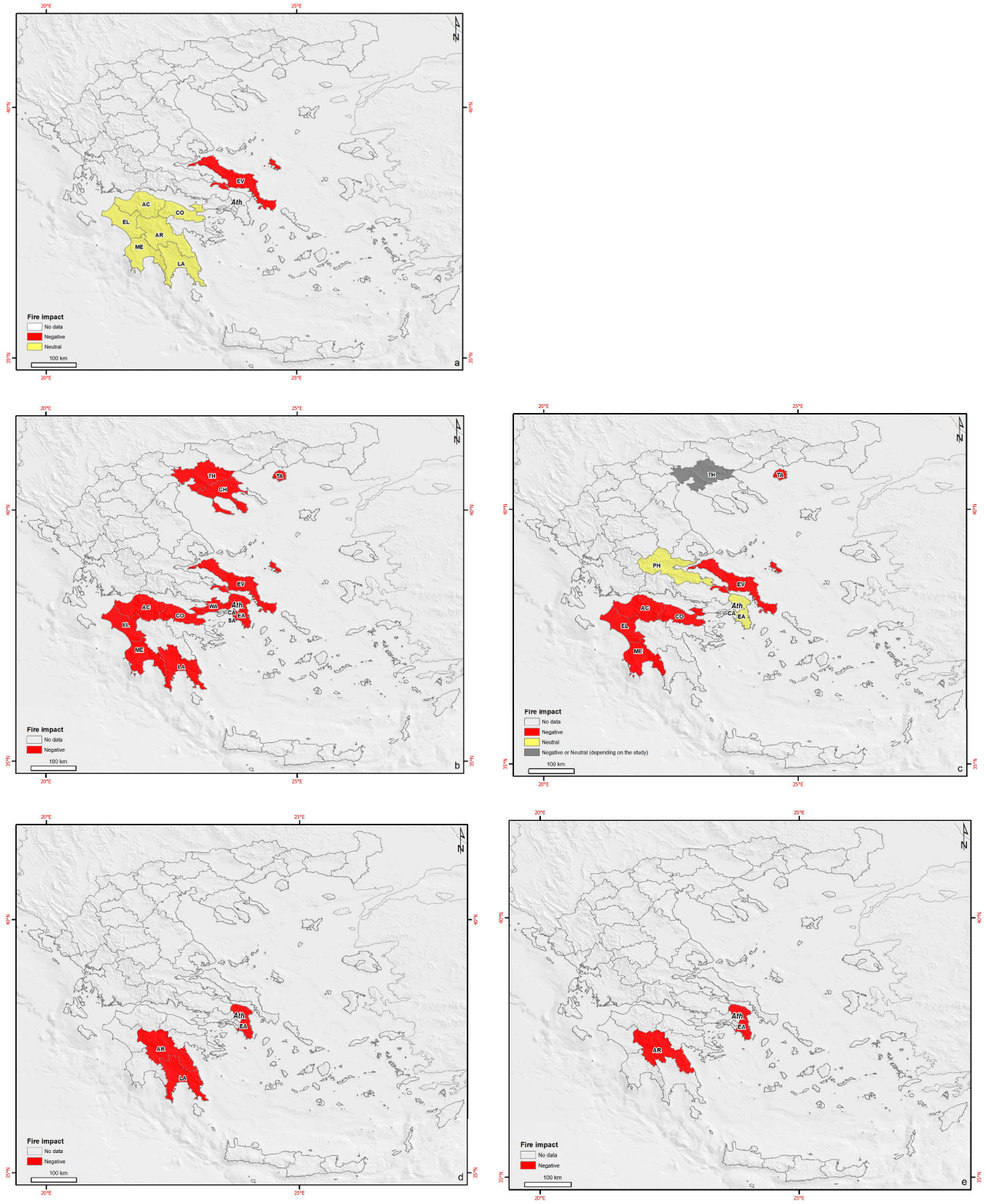


Figure 6. *Cont.*

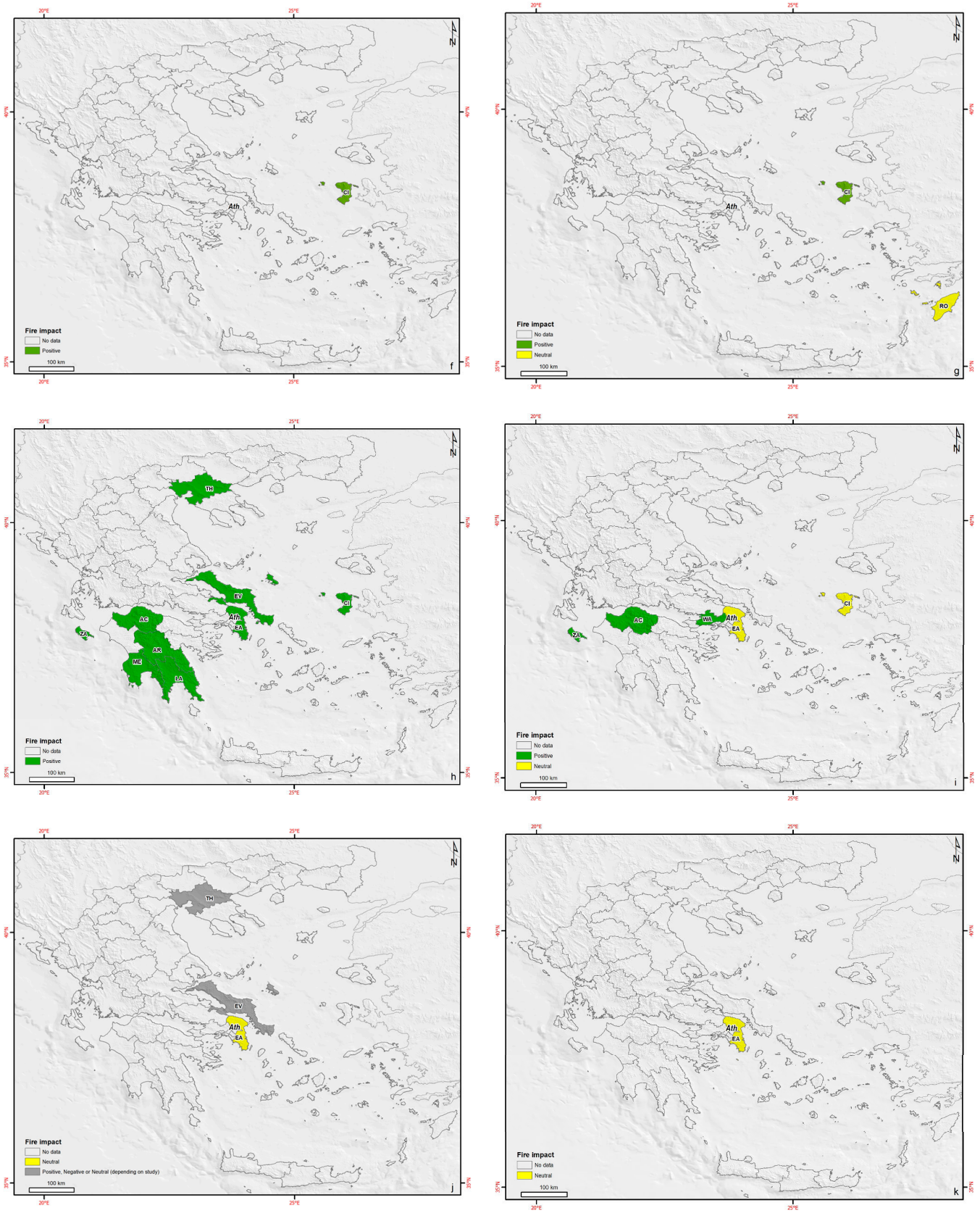


Figure 6. *Cont.*

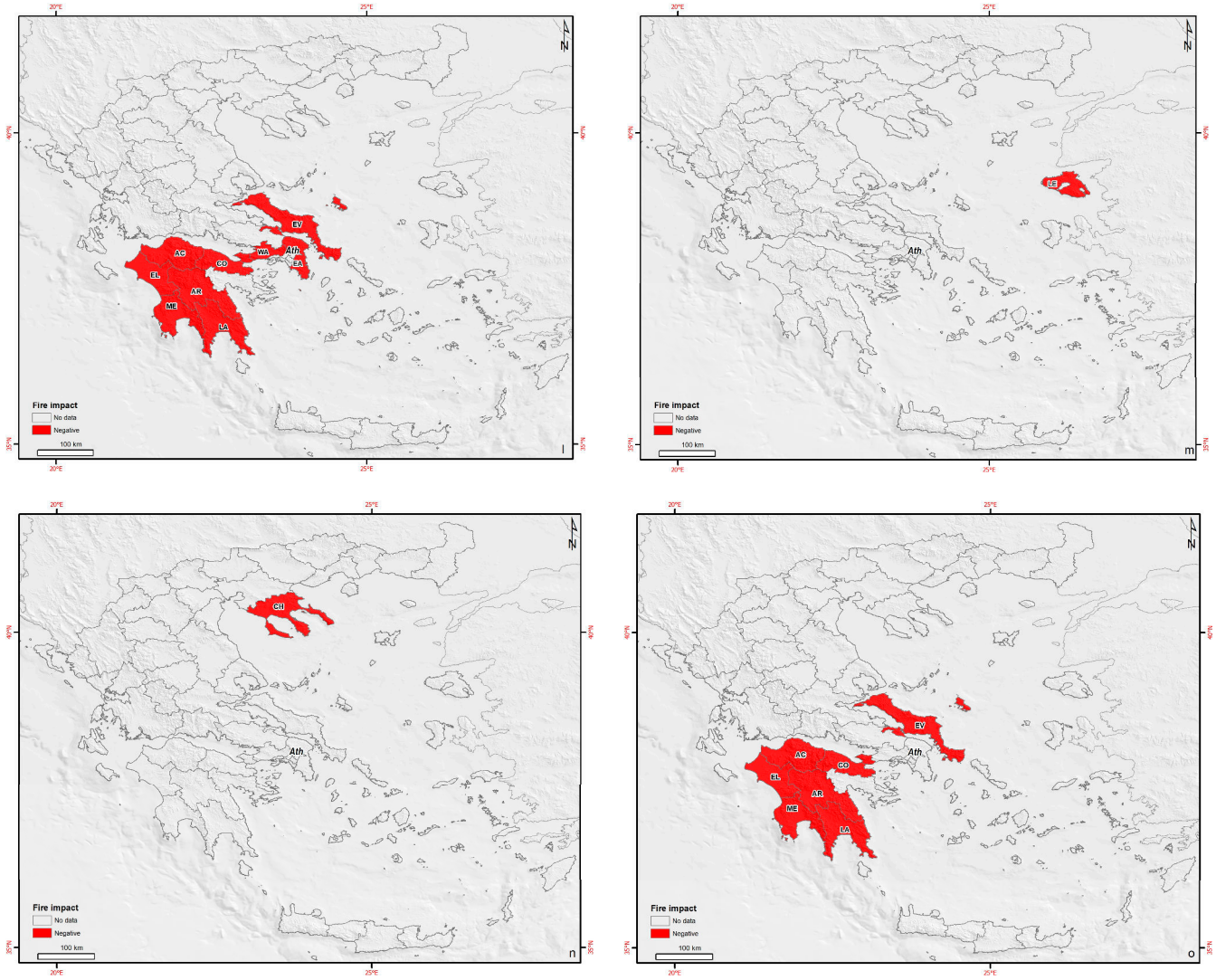


Figure 6. Geospatial distribution of the impacts of fire on ecosystem services in Greece, by regional unit and time span since last fire (short-term and medium-term effects are shown). (**a**) Provisioning services, short-term; (**b**) control of erosion rates, short-term; (**c**) control of erosion rates, medium-term; (**d**) regulation of the hydrologic cycle, medium-term; (**e**) regulation of the hydrologic cycle, medium-term; (**f**) pollination, short-term; (**g**) pollination, medium-term; (**h**) biodiversity, short-term; (**i**) biodiversity; medium-term; (**j**) regulation of soil quality, short-term; (**k**) regulation of soil quality, medium-term; (**l**) atmospheric composition and climate regulation, short-term; (**m**) atmospheric composition and climate regulation, medium-term; (**n**) cultural services, short-term; (**o**) cultural services, medium-term. Abbreviations for regional units are as follows: AC, Achaea; AR, Arcadia; CA, Central Athens; CH, Chalkidiki; CI, Chios; CO, Corinthia; EA, East Attica; EL, Elis; EV, Euboea; LA, Laconia; LE, Lesbos; ME, Messenia; PH, Phthiotis; RO, Rhodes; SA, South Athens; TA, Thasos; TH, Thessaloniki; WA, West Attica; ZA, Zakynthos. Ath stands for the city of Athens.

4. Discussion

4.1. Current Status of Research on the Impact of Fires on Forest Ecosystem Services in Greece

To the best of our knowledge, this is the first study on the impact of fire on ecosystem services in Greece. Based on the data collected and analyzed, the number of studies that examined the effects of wildfires on ecosystem services in Greece shows an increasing trend in the last 26 years (1 January 1997–10 March 2024), which is consistent with the findings of Roces-Díaz et al. [43], who conducted a similar study on a global scale and found an increase in the number of published articles during the period of 1993–2020, with the

highest representation of studies from the US, Spain, Canada, and Australia. The variation in the number of studies during this period examined is also consistent with the increase in research in the field of ecosystem services globally [25,26] and in Greece in particular [71] and with the number of fire incidents and burned area during the period of 1997–2024 in Greece [32,33].

The large representation of thermo-Mediterranean forests in this study (79% of all entries) may be explained by the fact that they are ecosystems with a high risk of exposure to fire and represent a large percentage of the burned area in Greece [72,73]. Also, due to their greater proximity to urban areas, they are more likely to be exposed to fires of anthropogenic origin, they are easier to access for studying the effects of fire, and fires in these areas have more direct socioeconomic impacts on humans. For research conducted on a global scale, Roces-Díaz et al. [43] found an over-representation of studies on the impact of fire on temperate and Mediterranean forests, while there was a lack of research on other types of ecosystems such as tropical and subtropical forests, despite the fact that fire is an important ecological disturbance in these ecosystems as well.

According to Acharya et al. [25], regulating services appear to be at the center of studies on forest ecosystem services. In the present work, most studies examined the impact of fire on regulating and maintenance services (94% of all entries), while very few focused on provisioning and cultural services. Among the regulating services, the ones mostly studied are the control of erosion rates and biodiversity (Figure 2). In the work of Roces-Diaz et al. [43], most entries (>90%) focused on the effects of fire on soil and water-related ecosystem services (soil fertility, erosion control, and water quality), indicating the growing interest of the scientific community in the interactions among fire, water, and soil [44,47,74]. Although the classification of ecosystem services used in this study differed to some extent from that of Roces-Diaz et al. [43], the results show similarities with those of the above study, especially regarding the control of erosion rates, as a significant increase in the number of studies of this ecosystem service has been observed in recent years. Another common element is the low representation of studies of fire impacts on cultural services, which is consistent with findings of other works that show limited studies on cultural ecosystem services, in general [25,43,71,75].

Regarding the time since the last fire, the largest proportion of entries (65%) considered short-term effects (Figure 3b). One can expect that the maximum number of impacts will be observed in the first period after the fire, and that, as time since the last fire increases and vegetation gradually recovers, the effects of fire will become milder. However, several studies also explored the longer-term impacts of fire and examined ecosystem services for a period longer than two years after the fire event (27% for 3–20 years after the fire and 8% for more than 20 years after the fire, Figure 3b). Additionally, 23 studies examined the same parameter at different time spans after the fire, something which shows the researchers' interest in post-fire restoration of ecosystem services.

Some studies examined broad areas or even the whole country, while others were limited to small areas. Also, several studies examined fires that affected protected areas, such as the Parnitha National Park, but in other studies that examined broader areas, no separate reference was made to protected areas. The geospatial representation of the results shows that for provisioning and cultural services, the regional units mostly studied were Euboea (fires of 2007 and 2021) and all the regional units of Peloponnese that were affected by the fires of 2007 (Corinthia, Achaea, Elis, Messenia, Laconia, and Arcadia), while for the regulating services for which the number of studies is higher, the regional units where most studies have been conducted are those of the Attica region (East Attica and West Attica), and of Euboea, and Laconia regional units (Figure 4). The megafires of 2007 and 2021 were examined in many of these studies, while the studies in Attica focus on fires that affected

the nearby mountains of Parnitha, Penteli, Hymettus, and Geraneia. Considering the fire statistics for Greece, the magnitude of the impact of fires in the years 2007, 2021, and 2023 is very high, in terms of both burned area and number of fire incidents [32]. The region of Attica has the largest ratio of burned area over its entire area, something that explains the high number of studies in Attica [32]. The recent study by Arianoutsou et al. [72] characterizes Attica as a hot spot of forest fires, with fires becoming larger in the period 1977–2024, while climate-related parameters, such as lower humidity, are increasing the fire risk.

4.2. Impacts of Fires on Forest Ecosystem Services in Greece

In this study, the negative effects of fire prevail when considering provisioning, cultural, and most regulating services examined (control of erosion rates, regulation of the hydrologic cycle, atmospheric composition, and climate regulation), while in the case of biodiversity and pollination, more positive effects were observed (Figure 5). It should be noted that the percentage of all entries showing negative effects of fire on ecosystem services (46.3%) vs. those showing positive effects (27.4%) recorded in our work are very similar to the respective percentages found by Roces-Díaz et al. [43]. The review of the latter, based on 207 studies on a global scale, found that most impacts of fire on forest ecosystem services were negative (46.6% of total effects), while a significant percentage of positive impacts was also recorded (28.5%). According to the same review [43], the prevalence of the perception that fires are a major socio-economic and environmental risk is very likely to have led to more research focusing on the negative rather than the potential positive short-term effects of fire and there is also a tendency to publish studies that show significant, rather than minor or no effects.

More specifically, the negative impact of fire on provisioning services related to biomass found in this study [76–80] is consistent with the findings of Roces-Díaz et al. [43] and Silvestro et al. [45], while Taboada et al. [81] found a positive impact of fire on the production of honey and resin. Roces-Díaz et al. [43] counted the increase in runoff following fire events as a positive effect on the provisioning of water; this service was not included as a separate category in our study. However, the serious disruptions in the hydrologic cycle following forest fire events are unlikely to prove beneficial for water provision in Greece, at a time when the Mediterranean region is experiencing changing precipitation regimes and is thus threatened with reduced water availability (leading to water scarcity and droughts), as well as flooding, due to climate change [62,82]. The negative effects of fire on cultural services recorded in our work [78,83–85] can be explained by the negative effect of fire on the esthetic value, the recreational value, but also the relational value (value that shapes the identity of those who live or work there) of the forest [86] and are in agreement with other studies [38,42,43,45,56,57]. Considering the increasing importance given nowadays to the link between natural environment and cultural heritage [87], it is expected that studies on these services will increase, something that is already beginning to be observed [25].

The negative impacts of fire on the control of erosion rates as well as on the disruption of the local hydrologic regime examined within this review [15,88–108] can be explained by the effect of fire on soil properties. The increase in soil hydrophobicity is followed by a reduction in soil infiltration and an increase in runoff and sediment yield, which further lead to higher flood risk at the downstream regions, especially when the relief is much smoother [44,54,74,109,110]. In their review study, Moazeni and Cerdà [74] concluded that the above short-term hydrologic changes affect groundwater recharge, baseflow, and water quality, with elevated levels of nutrients and other pollutants posing risks to human health and the aquatic ecosystem; these long-term effects need to be further studied. Additionally, deforestation and droughts, associated with reduced evapotranspiration, may trigger

ecosystem transitions and aggravate climate change at a regional level [111,112]. Other studies also report negative effects on the control of soil erosion rates [37,43,44,110], and on the regulation of the hydrologic cycle, with an increase in runoff and flood risk [36,46]. The largest number of entries on the impact of fire on the control of erosion rates in the present review refers to areas affected by the 2021 megafires in Attica, Euboea, and Peloponnese (Figure 6). Studies on water quality were not documented in our work, apart from the study by Megremi et al. [113] who refer to an indirect risk to downstream water quality from heavy metals found in soil leachate. Negative impacts of fire on water quality have been also reported by Nunes et al. [114] and Roces-Diaz et al. [43].

The positive and neutral effects of fire on pollination reported in our study [53,115–117] are correlated with the positive effects on plant diversity during the first post-fire season and are consistent with the findings of the review conducted by Carbone et al. [51], but not with those of Silvestro et al. [45], while the study of Roces-Díaz et al. [43] did not consider the evidence sufficient to reach conclusions. Based on the work of Petanidou et al. [118] in the Aegean region, the pollinator-plant relationship seems to depend on plant species richness, whereas the specificity of this relationship is influenced by climate, with temperature playing an important role in insect pollinator efficiency. In our work, we also investigated biodiversity, which is closely linked with all other ecosystem services and plays a key role in their provision [119]. Matossian et al. [120] studied different diversity indices in post-fire succession and came to different conclusions as to which one is most appropriate, depending on soil type. In the present review, papers that studied indicators such as species richness (α-diversity), Shannon index, or β-diversity were included and the results show more positive than neutral or negative effects on plant diversity in both thermo-Mediterranean and oro-Mediterranean ecosystems [40,41,50,121–132]. This could be explained by the resilience of many Mediterranean species, which have the ability to resprout or germinate from seeds after the fire. The openings created after the fire favor the establishment of new individuals [58] and ecological succession takes place, with most plant species being present in the post-fire community immediately after the fire [54]. The majority of studies on biodiversity refer to the region of Attica, whose mountainous areas have been significantly affected by fires in recent decades, and are characterized by an expansion of the urban–forest interface. The proportion of positive effects decreases as the post-fire period increases and the effects become more neutral in the long term. In contrast to plant diversity, the effects of fire on arthropod diversity were found to be negative. Based on Radea and Arianoutsou [129], high-severity fires in Mediterranean pine forests lead to a simplification of the soil community structure; El Khayati et al. [133] found differences in arthropod communities in burned and unburned areas in another area of the Mediterranean, and the response of each taxonomic group to fire appears to be related to specific functional characteristics such as habitat preference (open or forested landscapes) and ecological specialization (generalist or specialist species). Only one of the studies examined in the current review reported on the effects of fire on bird diversity and these were found to be neutral [128].

Regarding the effect of fire on the regulation of soil quality, results varied according to the parameter measured and the method used [50,77,113,125,131]. For example, while pH and decomposition rate did not appear to be significantly affected [50,125,131], changes in organic matter [125,131] and in the levels of metal ions were recorded [113]; the latter may pose a threat to water resources. Roces-Díaz et al. [43] also found mixed results on the effects of fire on soil fertility, with only 30–35% negative effects recorded. The magnitude of the effects on soil properties (soil nutrients, soil organic carbon) seems to be affected by fire severity [47,109,134–136], fire recurrence [47,109,125], and vegetation type [48,137], while

communities of soil organisms also appear to change with fire, and the effect also depends on fire regime (severity and frequency of the fires) [49,129].

Although the effects of fire severity and fire recurrence were not analyzed in the present study, fire severity has been found to affect post-fire recovery in a negative way both in Mediterranean and boreal forests [138,139]. On the other hand, recurrent fires with short time-intervals between fire incidents also impact regeneration [125,140] and affect multiple ecosystem services [141], while factors such as post-fire vegetation, topography and accessibility play a role in fire recurrence [142]. Poirazidis et al. [128] suggest that fires of intermediate frequency can play an important role in the maintenance of biodiversity in *Pinus halepensis* forests, according to the Intermediate Disturbance Hypothesis [143]. However, the changing fire regime will not impact all types of ecosystems in the same way and may lead to loss of resilience of the more vulnerable, not fire-adapted montane forest ecosystems [63].

The effects of fire on atmospheric composition and climate regulation were found to be negative overall in the short term and neutral in the long term [77,144–149]. The studies focused on the impacts of fire in larger areas of Greece, mostly affected by the 2007 and 2021 fires. The results are in agreement with the findings of Rocca et al. [37] on air quality and Molina et al. [150], Roces-Díaz et al. [43], and Silvestro et al. [45] on climate regulation.

Concerning the role of time since the last fire, all 23 studies that examined the effects of fire over different periods, including short-term, medium-term, and long-term, specifically focused on regulating services. Regarding the control of erosion rates, four studies found a decrease in the negative effects of fire over time [88,98,104,106], while three studies concluded that the negative effects continued several years after the fire [92,96,98]. Runoff was also found to increase in the first five years after the fire [89,99]. Regarding pollination services, two studies found no significant differences in the effects of fire on bee abundance and diversity 4 years and >20 years after the fire [115,116], while two other studies found that the positive effect of fire on pollinators decreases in the 2nd and 3rd year after the fire [53,117]. Regarding biodiversity, eight studies examined the effects of fire over time. Most of them observed an increase in plant diversity in the first 2–3 years after the fire [40,41,123,124,131,132], but this decreases with time [40,41,125], in accordance with the pattern of ecological succession in both thermo-Mediterranean and oro-Mediterranean forests. Variable effects on soil properties were observed during the first two years after the fire event by Spanos et al. [131], while the effects of fire on soil arthropod communities in the first 5 years after the fire were found to be negative [129]. According to Mayer et al. [136], the negative effects of fire on the soil's organic carbon persist for at least four decades after the fire event. Finally, negative effects on primary production (GPP, and thus carbon sequestration) were found 13 years after the fire in pine forests in the regional unit of Lesbos, and neutral effects in forests burned >40 years ago [149]. The results are in accordance with the findings of Roces-Diaz et al. [43] who also pointed out the role played by the time period after the last fire, particularly for the control of erosion rates and the regulation of the hydrologic cycle, where negative impacts are most pronounced in the first post-fire period. In the long-term studies included in their analysis, Roces-Dias et al. [43] found a negative impact of fire on climate regulation. The fact that fire effects decrease over the years in most of the studies reviewed in this paper can be explained by the gradual ecosystem restoration and recovery of ecosystem services, which may take up to 30–40 years in thermo-Mediterranean pine forests and longer in mountainous Mediterranean pine forests [54].

As the vast majority of studies in this review examined thermo-Mediterranean forest ecosystems, it is difficult to formulate a general opinion on how the type of ecosystem determines fire impacts. In practice, only four studies (11% of all entries) examined ecosys-

tems with only oro-Mediterranean forests [40,90,121,123]; these were montane ecosystems in the regional units of East Attica, Laconia, Arcadia, and Messenia, and the results in terms of controlling erosion rates, runoff, and plant diversity do not seem to differ from those of other relevant studies. What one would expect is a slower restoration of ecosystem services in these ecosystems, due to their slower regeneration [40,54]. The study by Rocez-Dias et al. [43] found no significant differences in the effects of fire on different types of ecosystems studied.

5. Conclusions

The review process revealed common trends regarding the effects of fire on each ecosystem service, with a prevalence of negative effects, mainly on provisioning and cultural services, on the control of erosion rates, the regulation of the hydrologic cycle, the atmospheric composition, and climate regulation, but a significant percentage of positive effects were found as well, mainly on pollination and plant diversity, while time since the last fire seems to moderate the effects. The studies reviewed here reported on different areas of Greece, with different types of ecosystems, with different pressures and fire history, and with a large variation in the area burned. They assessed different parameters/indicators for a given ecosystem service and used different methods to assess them. The spatial analysis of the studies indicated the regional units in which ecosystem services have been most affected by fire, with the region of Attica (regional units of East Attica and West Attica) showing the largest number of impacts, and the fires of 2007 and 2021 having produced the most devastating effects. The studies that were conducted in the same area, affected by the same fire incident, and measured a similar parameter, came up with similar results.

Fire disrupts ecosystem functions, affecting ecosystem services and social systems. Ecosystems with fire resilience characteristics will regain their functions over time, while vulnerable ecosystems may be degraded, if no actions for their conservation, fire management and post-fire recovery are taken. In the era of climate change and changing fire regimes, there is a need to further study the resilience of ecosystems to fire and to take measures to protect them in order to preserve their valuable biodiversity and the benefits it provides to humans.

Considering the lessons learned from the present study and the importance of the particular field of research as a source of insight for forest protection and management, the following recommendations could be made. In the area of research: (a) more economic valuation studies on ecosystem services should be conducted, as these will help better understand the losses due to fire; (b) more studies should be conducted on the impact of fire on provisioning and cultural services, to fill the gaps in these fields, as well as on the impact of fire on forests of high conservation value, such as those of protected areas, and the most vulnerable to fire mountainous forest ecosystems; (c) more studies are needed to understand the role of biodiversity in contributing to forest resilience and ecosystem services [151]. Research priorities should consider all stakeholders, potential funding sources, and regional needs. More interdisciplinary studies are needed, and there is a priority for funding research that is aligned with biodiversity conservation needs [152]. In the field of management, there should certainly be a greater focus on prevention, with more fire risk studies and identification of “hot spots” prone to fire, examination of different management scenarios, followed by prioritization of actions and appropriate prevention measures to improve ecosystem resilience. Management should aim at maintaining ecosystem characteristics that offer regulation against fire [153] and follow an “adaptation-first” model in which the adaptation of the forest should be the first priority [154]. In the field of communication, the following are important: (a) to give sufficient emphasis on the value and benefits of forest ecosystems for humans, to highlight their importance for achieving the goals of sustainable

development and human well-being [155], and (b) to create synergies with cultural agencies and to link the cultural heritage of a place with its natural capital, as cultural services may be more valued by the general public.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/f16030533/s1, Figure S1. PRISMA flow diagram for the review process. Figure S2. Number of entries per ecosystem service and ecosystem type; Figure S3. Number of entries per ecosystem service and time since last fire; Figure S4. The impact of fire on provisioning services. Number of entries vs. time since last fire; Figure S5. The impact of fire on the control of erosion rates. Number of entries vs. time since last fire; Figure S6. The impact of fire on the regulation of the hydrologic cycle. Number of entries vs. time since last fire; Figure S7. The impact of fire on pollination. Number of entries vs. time since last fire; Figure S8. The impact of fire on biodiversity, for three categories of organisms; Figure S9. The impact of fire on plant diversity. Number of entries per ecosystem type and time since last fire. TM, thermo-Mediterranean; OM, oro-Mediterranean; Figure S10. The impact of fire on regulation of soil quality. Number of entries vs. time since last fire; Figure S11. The impact of fire on atmospheric composition and climate regulation. Number of entries vs. time since last fire; Figure S12. The impact of fire on cultural services. Number of entries vs. time since last fire; Table S1. Number of studies and entries per major ecosystem services category; Table S2. Parameters measured and number of entries per parameter for each ecosystem category.

Author Contributions: Conceptualization, M.A. and A.M.; methodology, M.A., A.M. and A.C.; software, I.K., E.V. and A.M., validation; formal analysis, A.M.; investigation, A.M.; data curation, A.M.; writing—original draft preparation A.M.; writing—review and editing A.M., M.A., A.C. and I.K.; visualization, A.M., E.V. and I.K.; supervision, M.A.; project administration, M.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Data not included in the article or the Supplementary Materials can be requested from the authors.

Acknowledgments: This work was conducted within the framework of A.M.'s thesis for the MSc program in Ecology and Biodiversity Management, offered by the Faculty of Biology, National and Kapodistrian University of Athens. A.M. acknowledges Deree-The American College of Greece for being granted an AEGEAS CNPC Faculty Development Fellowship for academic year 2022–2023, to work on this project. The authors greatly acknowledge the three anonymous reviewers for their valuable comments that helped improve the manuscript.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Intergovernmental Panel On Climate Change (IPCC). Annex VII: Glossary. In *Climate Change 2021—The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*; Cambridge University Press: Cambridge, UK, 2023; pp. 2215–2256; ISBN 978-1-00-915789-6.
2. Harris, N.L.; Gibbs, D.A.; Baccini, A.; Birdsey, R.A.; De Bruin, S.; Farina, M.; Fatoyinbo, L.; Hansen, M.C.; Herold, M.; Houghton, R.A.; et al. Global Maps of Twenty-First Century Forest Carbon Fluxes. *Nat. Clim. Change* **2021**, *11*, 234–240. [CrossRef]
3. Pan, Y.; Birdsey, R.A.; Phillips, O.L.; Houghton, R.A.; Fang, J.; Kauppi, P.E.; Keith, H.; Kurz, W.A.; Ito, A.; Lewis, S.L.; et al. The Enduring World Forest Carbon Sink. *Nature* **2024**, *631*, 563–569. [CrossRef]
4. Jenkins, M.; Schaap, B. Forest Ecosystem Services. Untapped Potential: Forest Ecosystem Services for Achieving SDG 15 UNFF13 Background Analytical Study. 2018. Available online: https://www.forest-trends.org/publications/untapped-potential-forest-ecosystem-services-for-achieving-sdg-15/ (accessed on 11 March 2025).
5. FAO; Plan Bleu. *State of Mediterranean Forests 2018*; Food and Agriculture Organization of the United Nations, Rome and Plan Bleu: Marseille, France, 2018.
6. United Nations Department of Economic and Social Affairs. The Sustainable Development Goals Report 2023: Special Edition; United Nations: 2023; ISBN 978-92-1-002491-4. Available online: https://www.un-ilibrary.org/content/books/9789210024914/read (accessed on 11 March 2025). [CrossRef]

7. Altman, J.; Fibich, P.; Trotsiuk, V.; Altmanova, N. Global Pattern of Forest Disturbances and Its Shift under Climate Change. *Sci. Total Environ.* **2024**, *915*, 170117. [CrossRef] [PubMed]
8. Ke, P.; Ciais, P.; Sitch, S.; Li, W.; Bastos, A.; Liu, Z.; Xu, Y.; Gui, X.; Bian, J.; Goll, D.S.; et al. Low Latency Carbon Budget Analysis Reveals a Large Decline of the Land Carbon Sink in 2023. *Natl. Sci. Rev.* **2024**, *11*, nwae367. [CrossRef] [PubMed]
9. Yuan, X.; Chen, X.; Ochege, F.U.; Hamdi, R.; Tabari, H.; Li, B.; He, B.; Zhang, C.; De Maeyer, P.; Luo, G. Weakening of Global Terrestrial Carbon Sequestration Capacity under Increasing Intensity of Warm Extremes. *Nat. Ecol. Evol.* **2024**, *9*, 124–133. [CrossRef]
10. Jones, M.W.; Veraverbeke, S.; Andela, N.; Doerr, S.H.; Kolden, C.; Mataveli, G.; Pettinari, M.L.; Le Quéré, C.; Rosan, T.M.; Van Der Werf, G.R.; et al. Global Rise in Forest Fire Emissions Linked to Climate Change in the Extratropics. *Science* **2024**, *386*, eadl5889. [CrossRef] [PubMed]
11. Zheng, B.; Ciais, P.; Chevallier, F.; Chuvieco, E.; Chen, Y.; Yang, H. Increasing Forest Fire Emissions despite the Decline in Global Burned Area. *Sci. Adv.* **2021**, *7*, eabh2646. [CrossRef] [PubMed]
12. Hyyrynen, M.; Ollikainen, M.; Seppälä, J. European Forest Sinks and Climate Targets: Past Trends, Main Drivers, and Future Forecasts. *Eur. J. For. Res.* **2023**, *142*, 1207–1224. [CrossRef]
13. García-Ruiz, J.M.; Nadal-Romero, E.; Lana-Renault, N.; Beguería, S. Erosion in Mediterranean Landscapes: Changes and Future Challenges. *Geomorphology* **2013**, *198*, 20–36. [CrossRef]
14. Morán-Ordóñez, A.; Duane, A.; Gil-Tena, A.; De Cáceres, M.; Aquilué, N.; Guerra, C.A.; Geijzendorffer, I.R.; Fortin, M.; Brotons, L. Future Impact of Climate Extremes in the Mediterranean: Soil Erosion Projections When Fire and Extreme Rainfall Meet. *Land. Degrad. Dev.* **2020**, *31*, 3040–3054. [CrossRef]
15. Evelpidou, N.; Tzouxanioti, M.; Gavalas, T.; Spyrou, E.; Saitis, G.; Petropoulos, A.; Karkani, A. Assessment of Fire Effects on Surface Runoff Erosion Susceptibility: The Case of the Summer 2021 Forest Fires in Greece. *Land* **2021**, *11*, 21. [CrossRef]
16. Perea-Moreno, A.-J.; Perea-Moreno, M.-Á.; Hernandez-Escobedo, Q.; Manzano-Agugliaro, F. Towards Forest Sustainability in Mediterranean Countries Using Biomass as Fuel for Heating. *J. Clean. Prod.* **2017**, *156*, 624–634. [CrossRef]
17. Bloomer, L.; Dreyfus, G.; Ferris, T.; Zaelke, D.; Schiff, C. A Call to Stop Burning Trees in the Name of Climate Mitigation. *Vt. J. Env't L.* **2021**, *23*, 93.
18. Sterman, J.; Moomaw, W.; Rooney-Varga, J.N.; Siegel, L. Does Wood Bioenergy Help or Harm the Climate? *Bull. At. Sci.* **2022**, *78*, 128–138. [CrossRef]
19. Zauli-Sajani, S.; Thunis, P.; Pisoni, E.; Bessagnet, B.; Monforti-Ferrario, F.; De Meij, A.; Pekar, F.; Vignati, E. Reducing Biomass Burning Is Key to Decrease PM2.5 Exposure in European Cities. *Sci. Rep.* **2024**, *14*, 10210. [CrossRef] [PubMed]
20. Paraskevopoulou, D.; Kaskaoutis, D.G.; Grivas, G.; Bikkina, S.; Tsagkaraki, M.; Vrettou, I.M.; Tavernaraki, K.; Papoutsidaki, K.; Stavroulas, I.; Liakakou, E.; et al. Brown Carbon Absorption and Radiative Effects under Intense Residential Wood Burning Conditions in Southeastern Europe: New Insights into the Abundance and Absorptivity of Methanol-Soluble Organic Aerosols. *Sci. Total Environ.* **2023**, *860*, 160434. [CrossRef]
21. de-Miguel, S.; Pukkala, T.; Yeşil, A. Integrating Pine Honeydew Honey Production into Forest Management Optimization. *Eur. J. For. Res.* **2014**, *133*, 423–432. [CrossRef]
22. Mutke, S.; Calama, R.; González-Martínez, S.C.; Montero, G.; Gordo, F.J.; Bono, D.; Gil, L. Mediterranean Stone Pine: Botany and Horticulture. In *Horticultural Reviews*; Janick, J., Ed.; Wiley: Hoboken, NJ, USA, 2011; pp. 153–201; ISBN 978-1-118-09678-9.
23. Torres, I.; Moreno, J.M.; Morales-Molino, C.; Arianoutsou, M. Ecosystem Services Provided by Pine Forests. In *Pines and Their Mixed Forest Ecosystems in the Mediterranean Basin*; Ne'eman, G., Osem, Y., Eds.; Managing Forest Ecosystems; Springer International Publishing: Cham, Switzerland, 2021; Volume 38, pp. 617–629; ISBN 978-3-030-63624-1.
24. Guerra, C.A.; Maes, J.; Geijzendorffer, I.; Metzger, M.J. An Assessment of Soil Erosion Prevention by Vegetation in Mediterranean Europe: Current Trends of Ecosystem Service Provision. *Ecol. Indic.* **2016**, *60*, 213–222. [CrossRef]
25. Acharya, R.P.; Maraseni, T.; Cockfield, G. Global Trend of Forest Ecosystem Services Valuation—An Analysis of Publications. *Ecosyst. Serv.* **2019**, *39*, 100979. [CrossRef]
26. Chen, S.; Chen, J.; Jiang, C.; Yao, R.T.; Xue, J.; Bai, Y.; Wang, H.; Jiang, C.; Wang, S.; Zhong, Y.; et al. Trends in Research on Forest Ecosystem Services in the Most Recent 20 Years: A Bibliometric Analysis. *Forests* **2022**, *13*, 1087. [CrossRef]
27. Keeley, J.E.; Pausas, J.G.; Rundel, P.W.; Bond, W.J.; Bradstock, R.A. Fire as an Evolutionary Pressure Shaping Plant Traits. *Trends Plant Sci.* **2011**, *16*, 406–411. [CrossRef]
28. Paula, S.; Arianoutsou, M.; Kazanis, D.; Tavsanoglu, Ç.; Lloret, F.; Buhk, C.; Ojeda, F.; Luna, B.; Moreno, J.M.; Rodrigo, A.; et al. Fire-related Traits for Plant Species of the Mediterranean Basin: Ecological Archives E090-094. *Ecology* **2009**, *90*, 1420. [CrossRef]
29. Pausas, J.G.; Bradstock, R.A.; Keith, D.A.; Keeley, J.E. Plant Functional Traits in Relation to Fire in Crown-Fire Ecosystems. *Ecology* **2004**, *85*, 1085–1100. [CrossRef]
30. Pausas, J.G.; Ribeiro, E. Fire and Plant Diversity at the Global Scale. *Glob. Ecol. Biogeogr.* **2017**, *26*, 889–897. [CrossRef]
31. Rundel, P.W.; Arroyo, M.T.K.; Cowling, R.M.; Keeley, J.E.; Lamont, B.B.; Pausas, J.G.; Vargas, P. Fire and Plant Diversification in Mediterranean-Climate Regions. *Front. Plant Sci.* **2018**, *9*, 851. [CrossRef] [PubMed]

32. European Forest Fire Information System EFFIS Annual Statistics for Greece. Available online: https://forest-fire.emergency.copernicus.eu/apps/effis.statistics/estimates (accessed on 22 January 2025).
33. Kontoes, C.; Girtsou, S.; Yfantidou, A.; Tsoutsos, M.C.; Kokkalidou, M.; Apostolakis, A.; Giannopoulos, G.; Bartsotas, N.; Stathopoulos, N. Longitudinal Imaging and Analysis of Burnt Areas in Greece using Satellite Remote Sensing. In *The Resilience of Greek Forest Ecosystems to Climate Change*; Committee on the Resilience of Greek Forest Ecosystems to Climate Change (CFC), Arianoutsou, M., Zerefos, C., Kalabokidis, K., Poupkou, A., Aravanopoulos, F., Eds.; Academy of Athens: Athens, Greece, 2023; pp. 224–254; ISBN 978-960-404-413-9.
34. Keeley, J.E.; Pausas, J.G. Distinguishing Disturbance from Perturbations in Fire-Prone Ecosystems. *Int. J. Wildland Fire* **2019**, *28*, 282. [CrossRef]
35. Bond, W.J.; Woodward, F.I.; Midgley, G.F. The Global Distribution of Ecosystems in a World without Fire. *New Phytol.* **2005**, *165*, 525–538. [CrossRef]
36. Robinne, F.-N.; Hallema, D.W.; Bladon, K.D.; Buttle, J.M. Wildfire Impacts on Hydrologic Ecosystem Services in North American High-Latitude Forests: A Scoping Review. *J. Hydrol.* **2020**, *581*, 124360. [CrossRef]
37. Rocca, M.E.; Brown, P.M.; MacDonald, L.H.; Carrico, C.M. Climate Change Impacts on Fire Regimes and Key Ecosystem Services in Rocky Mountain Forests. *For. Ecol. Manag.* **2014**, *327*, 290–305. [CrossRef]
38. Vukomanovic, J.; Steelman, T. A Systematic Review of Relationships Between Mountain Wildfire and Ecosystem Services. *Landsc. Ecol.* **2019**, *34*, 1179–1194. [CrossRef]
39. Alfaro-Sánchez, R.; Sánchez-Salguero, R.; De Las Heras, J.; Hernández-Tecles, E.; Moya, D.; López-Serrano, F.R. Vegetation Dynamics of Managed M Editerranean Forests 16 Yr after Large Fires in Southeastern S Pain. *Appl. Veg. Sci.* **2015**, *18*, 272–282. [CrossRef]
40. Christopoulou, A.; Kazanis, D.; Fyllas, N.; Arianoutsou, M. Post-Fire Recovery of Abies Cephalonica Forest Communities: The Case of Mt Parnitha National Park, Attica, Greece. *iForest* **2018**, *11*, 757–764. [CrossRef]
41. Kazanis, D.; Arianoutsou, M. Long-Term Post-Fire Vegetation Dynamics in Pinus Halepensis Forests of Central Greece: A Functional Group Approach. *Plant Ecol.* **2004**, *171*, 101–121. [CrossRef]
42. Mancilla-Ruiz, D.; Barrera, F.D.L.; González, S.; Huaico, A. The Effects of a Megafire on Ecosystem Services and the Pace of Landscape Recovery. *Land* **2021**, *10*, 1388. [CrossRef]
43. Roces-Díaz, J.V.; Santín, C.; Martínez-Vilalta, J.; Doerr, S.H. A Global Synthesis of Fire Effects on Ecosystem Services of Forests and Woodlands. *Front. Ecol. Environ.* **2022**, *20*, 170–178. [CrossRef]
44. Shakesby, R.A. Post-Wildfire Soil Erosion in the Mediterranean: Review and Future Research Directions. *Earth-Sci. Rev.* **2011**, *105*, 71–100. [CrossRef]
45. Silvestro, R.; Saulino, L.; Cavallo, C.; Allevato, E.; Pindozzi, S.; Cervelli, E.; Conti, P.; Mazzoleni, S.; Saracino, A. The Footprint of Wildfires on Mediterranean Forest Ecosystem Services in Vesuvius National Park. *Fire* **2021**, *4*, 95. [CrossRef]
46. Vieira, D.C.S.; Fernández, C.; Vega, J.A.; Keizer, J.J. Does Soil Burn Severity Affect the Post-Fire Runoff and Interrill Erosion Response? A Review Based on Meta-Analysis of Field Rainfall Simulation Data. *J. Hydrol.* **2015**, *523*, 452–464. [CrossRef]
47. Caon, L.; Vallejo, V.R.; Ritsema, C.J.; Geissen, V. Effects of Wildfire on Soil Nutrients in Mediterranean Ecosystems. *Earth-Sci. Rev.* **2014**, *139*, 47–58. [CrossRef]
48. Fernández-García, V.; Marcos, E.; Huerta, S.; Calvo, L. Soil-Vegetation Relationships in Mediterranean Forests after Fire. *For. Ecosyst.* **2021**, *8*, 18. [CrossRef]
49. Mikita-Barbato, R.A.; Kelly, J.J.; Tate, R.L. Wildfire Effects on the Properties and Microbial Community Structure of Organic Horizon Soils in the New Jersey Pinelands. *Soil. Biol. Biochem.* **2015**, *86*, 67–76. [CrossRef]
50. Radea, C.; Arianoutsou, M. Cellulose Decomposition Rates and Soil Arthropod Community in a Pinus Halepensis Mill. Forest of Greece after a Wildfire. *Eur. J. Soil. Biol.* **2000**, *36*, 57–64. [CrossRef]
51. Carbone, L.M.; Tavella, J.; Pausas, J.G.; Aguilar, R. A Global Synthesis of Fire Effects on Pollinators. *Glob. Ecol. Biogeogr.* **2019**, *28*, 1487–1498. [CrossRef]
52. Lazarina, M.; Devalez, J.; Neokosmidis, L.; Sgardelis, S.P.; Kallimanis, A.S.; Tscheulin, T.; Tsalkatis, P.; Kourtidou, M.; Mizerakis, V.; Nakas, G.; et al. Moderate Fire Severity Is Best for the Diversity of Most of the Pollinator Guilds in Mediterranean Pine Forests. *Ecology* **2019**, *100*, e02615. [CrossRef] [PubMed]
53. Nakas, G.; Kantsa, A.; Vujić, A.; Mescher, M.C.; De Moraes, C.M.; Petanidou, T. Recent Fire in a Mediterranean Ecosystem Strengthens Hoverfly Populations and Their Interaction Networks with Plants. *Ecol. Evol.* **2023**, *13*, e9803. [CrossRef]
54. Arianoutsou-Faraggitaki, M.; Kazanis, D. The ecological role of fire in the terrestrial ecosystems of Greece. In *The Forest: An Integrated Approach*; Papageorgiou, A.C., Karetsos, G., Katsadorakis, G., Eds.; WWF Greece: Athens, Greece, 2012; pp. 103–116.
55. Kazanis, D.; Spatharis, S.; Kokkoris, G.D.; Dimitrakopoulos, P.G.; Arianoutsou, M. Drivers of Pinus Halepensis Plant Community Structure across a Post-Fire Chronosequence. *Fire* **2024**, *7*, 331. [CrossRef]
56. Pereira, P.; Bogunovic, I.; Zhao, W.; Barcelo, D. Short-Term Effect of Wildfires and Prescribed Fires on Ecosystem Services. *Curr. Opin. Environ. Sci. Health* **2021**, *22*, 100266. [CrossRef]

57. Rosselló, J.; Becken, S.; Santana-Gallego, M. The Effects of Natural Disasters on International Tourism: A Global Analysis. *Tour. Manag.* **2020**, *79*, 104080. [CrossRef]
58. Pausas, J.G.; Keeley, J.E. Wildfires as an Ecosystem Service. *Front. Ecol. Environ.* **2019**, *17*, 289–295. [CrossRef]
59. MedECC Climate and Environmental Change in the Mediterranean Basin—Current Situation and Risks for the Future. In *First Mediterranean Assessment Report*; Cramer, W., Guiot, J., Marini, K., Eds.; Union for the Mediterranean, Plan Bleu, UNEP/MAP: Marseille, France, 2020; ISBN 978-2-9577416-0-1.
60. Arianoutsou, M.; Aravanopoulos, F.; Dimopoulos, P.; Kokkoris, I.; Mallinis, G.; Pafilis, P.; Christopoulou, A. Biodiversity and Ecosystem Services of Greek Forests. In *The Resilience of Greek Forest Ecosystems to Climate Change*; Committee on the Resilience of Greek Forest Ecosystems to Climate Change (CFC), Arianoutsou, M., Zerefos, C., Kalabokidis, K., Poupkou, A., Aravanopoulos, F., Eds.; Academy of Athens: Athens, Greece, 2023; pp. 165–199; ISBN 978-960-404-413-9.
61. Kokkoris, I.; Dimopoulos, P.; Xystrakis, F.; Tsiripidis, I. National Scale Ecosystem Condition Assessment with Emphasis on Forest Types in Greece. *One Ecosyst.* **2018**, *3*, e25434. [CrossRef]
62. Intergovernmental Panel On Climate Change (IPCC). *Climate Change 2022—Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, 1st ed.; Cambridge University Press: Cambridge, UK, 2023; ISBN 978-1-00-932584-4.
63. Arianoutsou, M.; Aravanopoulos, F.; Fyllas, N.M.; Kapsomenakis, I.; Zerefos, C. Climate Change and Resilience of Forest Ecosystems. In *The Resilience of Greek Forest Ecosystems to Climate Change*; Committee on the Resilience of Greek Forest Ecosystems to Climate Change (CFC), Arianoutsou, M., Zerefos, C., Kalabokidis, K., Poupkou, A., Aravanopoulos, F., Eds.; Academy of Athens: Athens, Greece, 2023; pp. 71–82; ISBN 978-960-404-413-9.
64. Fyllas, N.M.; Koufaki, T.; Sazeides, C.I.; Spyroglou, G.; Theodorou, K. Potential Impacts of Climate Change on the Habitat Suitability of the Dominant Tree Species in Greece. *Plants* **2022**, *11*, 1616. [CrossRef] [PubMed]
65. Brandes, R.; Christopoulou, A. Long-Term Forest Dynamics of Oromediterranean Fir Forests in Greece. *Seefor* **2020**, *11*, 71–84. [CrossRef]
66. Koulelis, P.P.; Proutsos, N.; Solomou, A.D.; Avramidou, E.V.; Malliarou, E.; Athanasiou, M.; Xanthopoulos, G.; Petrakis, P.V. Effects of Climate Change on Greek Forests: A Review. *Atmosphere* **2023**, *14*, 1155. [CrossRef]
67. Lecina-Diaz, J.; Alvarez, A.; De Cáceres, M.; Herrando, S.; Vayreda, J.; Retana, J. Are Protected Areas Preserving Ecosystem Services and Biodiversity? Insights from Mediterranean Forests and Shrublands. *Landsc. Ecol.* **2019**, *34*, 2307–2321. [CrossRef]
68. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ* **2021**, *372*, n71. [CrossRef]
69. Arianoutsou, M. Aspects of Demography in Post-Fire Mediterranean Plant Communities of Greece. In *Landscape Disturbance and Biodiversity in Mediterranean-Type Ecosystems*; Rundel, P.W., Montenegro, G., Jaksic, F.M., Eds.; Ecological Studies; Springer Berlin Heidelberg: Berlin/Heidelberg, Germany, 1998; Volume 136, pp. 273–295; ISBN 978-3-642-08416-4.
70. Tapias, R.; Climent, J.; Pardos, J.A.; Gil, L. Life Histories of Mediterranean Pines. *Plant Ecol.* **2004**, *171*, 53–68. [CrossRef]
71. Dimopoulos, P.; Drakou, E.; Kokkoris, I.; Katsanevakis, S.; Kallimanis, A.; Tsiafouli, M.; Bormpoudakis, D.; Kormas, K.; Arends, J. The Need for the Implementation of an Ecosystem Services Assessment in Greece: Drafting the National Agenda. *One Ecosyst.* **2017**, *2*, e13714. [CrossRef]
72. Arianoutsou, M.; Athanasakis, G.; Kazanis, D.; Christopoulou, A. Attica: A Hot Spot for Forest Fires in Greece. *Fire* **2024**, *7*, 467. [CrossRef]
73. Moreno, J.M.; Morales-Molino, C.; Torres, I.; Arianoutsou, M. Fire in Mediterranean Pine Forests: Past, Present and Future. In *Pines and Their Mixed Forest Ecosystems in the Mediterranean Basin*; Ne'eman, G., Osem, Y., Eds.; Managing Forest Ecosystems; Springer International Publishing: Cham, Switzerland, 2021; Volume 38, pp. 421–456; ISBN 978-3-030-63624-1.
74. Moazeni, S.; Cerdà, A. The Impacts of Forest Fires on Watershed Hydrological Response. A Review. *Trees For. People* **2024**, *18*, 100707. [CrossRef]
75. Satz, D.; Gould, R.K.; Chan, K.M.A.; Guerry, A.; Norton, B.; Satterfield, T.; Halpern, B.S.; Levine, J.; Woodside, U.; Hannahs, N.; et al. The Challenges of Incorporating Cultural Ecosystem Services into Environmental Assessment. *Ambio* **2013**, *42*, 675–684. [CrossRef]
76. Costopoulou, D.; Vassiliadou, I.; Chrysafidis, D.; Bergele, K.; Tzavara, E.; Tzamtzis, V.; Leondiadis, L. Determination of PCDD/F, Dioxin-like PCB and PAH Levels in Olive and Olive Oil Samples from Areas Affected by the Fires in Summer 2007 in Greece. *Chemosphere* **2010**, *79*, 285–291. [CrossRef] [PubMed]
77. Kalfas, D.; Kalogiannidis, S.; Chatzitheodoridis, F.; Margaritis, N. The Other Side of Fire in a Changing Environment: Evidence from a Mediterranean Country. *Fire* **2024**, *7*, 36. [CrossRef]
78. Palaiologou, P.; Kalabokidis, K.; Troumbis, A.; Day, M.A.; Nielsen-Pincus, M.; Ager, A.A. Socio-Ecological Perceptions of Wildfire Management and Effects in Greece. *Fire* **2021**, *4*, 18. [CrossRef]

79. Stougiannidou, D.; Zafeiriou, E.; Raftoyannis, Y. Forest Fires in Greece and Their Economic Impacts on Agriculture. *KSS* **2020**, 54–70. [CrossRef]
80. Tsioras, P.A.; Giamouki, C.; Tsaktsira, M.; Scaltsoyiannes, A. What the Fire Has Left Behind: Views and Perspectives of Resin Tappers in Central Greece. *Sustainability* **2023**, *15*, 9777. [CrossRef]
81. Taboada, A.; García-Llamas, P.; Fernández-Guisuraga, J.M.; Calvo, L. Wildfires Impact on Ecosystem Service Delivery in Fire-Prone Maritime Pine-Dominated Forests. *Ecosyst. Serv.* **2021**, *50*, 101334. [CrossRef]
82. Zittis, G.; Almazroui, M.; Alpert, P.; Ciais, P.; Cramer, W.; Dahdal, Y.; Fnais, M.; Francis, D.; Hadjinicolaou, P.; Howari, F.; et al. Climate Change and Weather Extremes in the Eastern Mediterranean and Middle East. *Rev. Geophys.* **2022**, *60*, e2021RG000762. [CrossRef]
83. Boustras, G.; Boukas, N. Forest Fires' Impact on Tourism Development: A Comparative Study of Greece and Cyprus. *Manag. Environ. Qual.* **2013**, *24*, 498–511. [CrossRef]
84. Mallinis, G.; Mitsopoulos, I.; Beltran, E.; Goldammer, J. Assessing Wildfire Risk in Cultural Heritage Properties Using High Spatial and Temporal Resolution Satellite Imagery and Spatially Explicit Fire Simulations: The Case of Holy Mount Athos, Greece. *Forests* **2016**, *7*, 46. [CrossRef]
85. Papaspyropoulos, K.; Sokos, C.; Birtsas, P. The Impacts of a Wildfire on Hunting Demand: A Case Study of a Mediterranean Ecosystem. *iForest* **2015**, *8*, 95–100. [CrossRef]
86. Orenstein, D.E. The Cultural Ecosystem Services of Mediterranean Pine Forests. In *Pines and Their Mixed Forest Ecosystems in the Mediterranean Basin*; Ne'eman, G., Osem, Y., Eds.; Managing Forest Ecosystems; Springer International Publishing: Cham, Switzerland, 2021; Volume 38, pp. 631–655; ISBN 978-3-030-63624-1.
87. Larsen, P.B.; Wijesuriya, G. Nature–Culture Interlinkages in World Heritage: Bridging the Gap. *Georg. Wright Forum* **2017**, *34*, 142–153.
88. Alexiou, S.; Papanikolaou, I.; Schneiderwind, S.; Kehrle, V.; Reicherter, K. Monitoring and Quantifying Soil Erosion and Sedimentation Rates in Centimeter Accuracy Using UAV-Photogrammetry, GNSS, and t-LiDAR in a Post-Fire Setting. *Remote Sens.* **2024**, *16*, 802. [CrossRef]
89. Batelis, S.C.; Nalbantis, I. A Multi-Model Multi-Scale Approach to Estimate the Impact of the 2007 Large-Scale Forest Fires in Peloponnese, Greece. *Water* **2022**, *14*, 3348. [CrossRef]
90. Blake, W.H.; Theocharopoulos, S.P.; Skoulikidis, N.; Clark, P.; Tountas, P.; Hartley, R.; Amaxidis, Y. Wildfire Impacts on Hillslope Sediment and Phosphorus Yields. *J. Soils Sediments* **2010**, *10*, 671–682. [CrossRef]
91. Deligiannakis, G.; Pallikarakis, A.; Papanikolaou, I.; Alexiou, S.; Reicherter, K. Detecting and Monitoring Early Post-Fire Sliding Phenomena Using UAV–SfM Photogrammetry and t-LiDAR-Derived Point Clouds. *Fire* **2021**, *4*, 87. [CrossRef]
92. Diakakis, M.; Nikolopoulos, E.I.; Mavroulis, S.; Vassilakis, E.; Korakaki, E. Observational Evidence on the Effects of Mega-Fires on the Frequency of Hydrogeomorphic Hazards. The Case of the Peloponnese Fires of 2007 in Greece. *Sci. Total Environ.* **2017**, *592*, 262–276. [CrossRef] [PubMed]
93. Efthimiou, N.; Psomiadis, E.; Panagos, P. Fire Severity and Soil Erosion Susceptibility Mapping Using Multi-Temporal Earth Observation Data: The Case of Mati Fatal Wildfire in Eastern Attica, Greece. *CATENA* **2020**, *187*, 104320. [CrossRef]
94. Karamesouti, M.; Petropoulos, G.P.; Papanikolaou, I.D.; Kairis, O.; Kosmas, K. Erosion Rate Predictions from PESERA and RUSLE at a Mediterranean Site before and after a Wildfire: Comparison & Implications. *Geoderma* **2016**, *261*, 44–58. [CrossRef]
95. Kastridis, A.; Stathis, D.; Sapountzis, M.; Theodosiou, G. Insect Outbreak and Long-Term Post-Fire Effects on Soil Erosion in Mediterranean Suburban Forest. *Land* **2022**, *11*, 911. [CrossRef]
96. Mallinis, G.; Gitas, I.Z.; Tasionas, G.; Maris, F. Multitemporal Monitoring of Land Degradation Risk Due to Soil Loss in a Fire-Prone Mediterranean Landscape Using Multi-Decadal Landsat Imagery. *Water Resour. Manag.* **2016**, *30*, 1255–1269. [CrossRef]
97. Mallinis, G.; Maris, F.; Kalinderis, I.; Koutsias, N. Assessment of Post-Fire Soil Erosion Risk in Fire-Affected Watersheds Using Remote Sensing and GIS. *GISci. Remote Sens.* **2009**, *46*, 388–410. [CrossRef]
98. Margiorou, S.; Kastridis, A.; Sapountzis, M. Pre/Post-Fire Soil Erosion and Evaluation of Check-Dams Effectiveness in Mediterranean Suburban Catchments Based on Field Measurements and Modeling. *Land* **2022**, *11*, 1705. [CrossRef]
99. Moussoulis, E.; Mallinis, G.; Koutsias, N.; Zacharias, I. Modelling Surface Runoff to Evaluate the Effects of Wildfires in Multiple Semi-arid, Shrubland-dominated Catchments. *Hydrol. Process.* **2015**, *29*, 4427–4441. [CrossRef]
100. Myronidis, D.I.; Emmanouloudis, D.A.; Mitsopoulos, I.A.; Riggos, E.E. Soil Erosion Potential after Fire and Rehabilitation Treatments in Greece. *Environ. Model. Assess.* **2010**, *15*, 239–250. [CrossRef]
101. Skilodimou, H.D.; Bathrellos, G.D.; Alexakis, D.E. Flood Hazard Assessment Mapping in Burned and Urban Areas. *Sustainability* **2021**, *13*, 4455. [CrossRef]
102. Soulis, K.X.; Generali, K.A.; Papadaki, C.; Theodoropoulos, C.; Psomiadis, E. Hydrological Response of Natural Mediterranean Watersheds to Forest Fires. *Hydrology* **2021**, *8*, 15. [CrossRef]

103. Stefanidis, S.; Alexandridis, V. Wildfire Effects on Soil Erosion and Hydrology in a Typical Medieterranean Wetland-Urban Interface: The Case of South Athens, Greece. In Proceedings of the Safe Thessaloniki 2022, Thessaloniki, Greece, 29 September–1 October 2022; pp. 383–386.
104. Stefanidis, S.; Alexandridis, V.; Spalevic, V.; Luiz Mincato, R. Wildfire Effects on Soil Erosion Dynamics: The Case of 2021 Megafires in Greece. *AgricultForest* **2022**, *68*, 49–63. [CrossRef]
105. Stefanidis, S.; Alexandridis, V.; Mallinis, G. A Cloud-Based Mapping Approach for Assessing Spatiotemporal Changes in Erosion Dynamics Due to Biotic and Abiotic Disturbances in a Mediterranean Peri-Urban Forest. *CATENA* **2022**, *218*, 106564. [CrossRef]
106. Tselka, I.; Krassakis, P.; Rentzelos, A.; Koukouzas, N.; Parcharidis, I. Assessing Post-Fire Effects on Soil Loss Combining Burn Severity and Advanced Erosion Modeling in Malesina, Central Greece. *Remote Sens.* **2021**, *13*, 5160. [CrossRef]
107. Valkanou, K.; Karymbalis, E.; Bathrellos, G.; Skilodimou, H.; Tsanakas, K.; Papanastassiou, D.; Gaki-Papanastassiou, K. Soil Loss Potential Assessment for Natural and Post-Fire Conditions in Evia Island, Greece. *Geosciences* **2022**, *12*, 367. [CrossRef]
108. Vassilakis, E.; Farangitakis, G.P.; Christopoulou, A.; Antoniou, V.; Arianoutsou, M. Assessment of Different Methods for the Quantification of Soil Erosion Risk after Fire: A Case Study from Geraneia Mt., Central Greece. In *Geophysical Research Abstracts 21*; EGU General Assembly: Vienna, Austria, 2019.
109. Francos, M.; Colino-Prieto, F.; Sánchez-García, C. How Mediterranean Ecosystem Deals with Wildfire Impact on Soil Ecosystem Services and Functions: A Review. *Land* **2024**, *13*, 407. [CrossRef]
110. Wittenberg, L. Post-Fire Soil Erosion—The Mediterranean Perception. In *Pines and Their Mixed Forest Ecosystems in the Mediterranean Basin*; Ne'eman, G., Osem, Y., Eds.; Managing Forest Ecosystems; Springer International Publishing: Cham, Switzerland, 2021; Volume 38, pp. 481–496; ISBN 978-3-030-63624-1.
111. Flores, B.M.; Montoya, E.; Sakschewski, B.; Nascimento, N.; Staal, A.; Betts, R.A.; Levis, C.; Lapola, D.M.; Esquível-Muelbert, A.; Jakovac, C.; et al. Critical Transitions in the Amazon Forest System. *Nature* **2024**, *626*, 555–564. [CrossRef] [PubMed]
112. Xu, X.; Zhang, X.; Riley, W.J.; Xue, Y.; Nobre, C.A.; Lovejoy, T.E.; Jia, G. Deforestation Triggering Irreversible Transition in Amazon Hydrological Cycle. *Environ. Res. Lett.* **2022**, *17*, 034037. [CrossRef]
113. Megremi, I.; Stathopoulou, E.; Vorris, E.; Kostakis, M.; Karavoltsos, S.; Thomaidis, N.; Vasilatos, C. Mediterranean Wildfires' Effect on Soil Quality and Properties: A Case from Northern Euboea, Greece. *Land* **2024**, *13*, 325. [CrossRef]
114. Nunes, J.P.; Carvalho-Santos, C.; Pastor, A. Impacts of Wildfires on Hydrological Ecosystem Services. *Proceedings* **2019**, *30*, 48.
115. Lazarina, M.; Sgardelis, S.P.; Tscheulin, T.; Devalez, J.; Mizerakis, V.; Kallimanis, A.S.; Papakonstantinou, S.; Kyriazis, T.; Petanidou, T. The Effect of Fire History in Shaping Diversity Patterns of Flower-Visiting Insects in Post-Fire Mediterranean Pine Forests. *Biodivers. Conserv.* **2017**, *26*, 115–131. [CrossRef]
116. Lazarina, M.; Sgardelis, S.P.; Tscheulin, T.; Kallimanis, A.S.; Devalez, J.; Petanidou, T. Bee Response to Fire Regimes in Mediterranean Pine Forests: The Role of Nesting Preference, Trophic Specialization, and Body Size. *Basic. Appl. Ecol.* **2016**, *17*, 308–320. [CrossRef]
117. Nakas, G.; Lazarina, M.; Chtenelli, T.; Sgardelis, S.; Dimitrakopoulos, P.G.; Devalez, J.; Petanidou, T. The Response to Fire by Two Eusocial Bee Species. *Ecol. Entomol.* **2024**, *49*, 41–53. [CrossRef]
118. Petanidou, T.; Kallimanis, A.S.; Lazarina, M.; Tscheulin, T.; Devalez, J.; Stefanaki, A.; Hanlidou, E.; Vujić, A.; Kaloveloni, A.; Sgardelis, S.P. Climate Drives Plant–Pollinator Interactions Even along Small-scale Climate Gradients: The Case of the Aegean. *Plant Biol. J.* **2018**, *20*, 176–183. [CrossRef]
119. Maes, J.; Liquete, C.; Teller, A.; Erhard, M.; Paracchini, M.L.; Barredo, J.I.; Grizzetti, B.; Cardoso, A.; Somma, F.; Petersen, J.-E.; et al. An Indicator Framework for Assessing Ecosystem Services in Support of the EU Biodiversity Strategy to 2020. *Ecosyst. Serv.* **2016**, *17*, 14–23. [CrossRef]
120. Matossian, A.D.; Matsinos, Y.G.; Konstantinidis, P.; Moustakas, A. Post-Fire Succession Indices Performance in a Mediterranean Ecosystem. *Stoch. Environ. Res. Risk Assess.* **2013**, *27*, 323–335. [CrossRef]
121. Arianoutsou, M.; Christopoulou, A.; Kazanis, D.; Tountas, T.; Ganou, E.; Bazos, I.; Kokkoris, Y. Effects of Fire on High Altitude Coniferous Forests of Greece. In Proceedings of the VI International Conference on Forest Fire Research 2010, Coimbra, Portugal, 15–18 November 2010.
122. Ganatsas, P.; Spanos, I.; Tsakaldimi, M.; Goudelis, G. Soil Nutrient, Woody Understory and Shoot and Root Growth Responses of Pinus Brutia Ten. Saplings to Fire. *Not. Bot. Hort. Agrobot. Cluj.* **2012**, *40*, 302. [CrossRef]
123. Georgiadis, T.; Dimitrelos, G. *Report on the Releves in Burnt Areas of the Habitat Type "Sub-Mediterranean Pine Forests with Endemic Black Pine" 9530 on Mount Parnonas (GR2520006)*; Greek Biotope-Wetland Centre: Thermi, Greece, 2010.
124. Goudelis, G.; Ganatsas, P.P.; Spanos, I.; Karpi, A. Effect of Repeated Fire on Plant Community Recovery in Penteli, Central Greece. In *Eco-and Ground Bio-Engineering: The Use of Vegetation to Improve Slope Stability*; Stokes, A., Spanos, I., Norris, J.E., Cammeraat, E., Eds.; Springer Netherlands: Dordrecht, The Netherlands, 2007; pp. 337–343; ISBN 978-1-4020-5592-8.
125. Goudelis, G.; Ganatsas, P.; Tsitsoni, T.; Spanos, Y.; Daskalakou, E. Effect of Two Successive Wildfires in Pinus Halepensis Stands of Central Greece. *Web Ecol.* **2008**, *8*, 30–34. [CrossRef]

126. Nakas, G.; Kougioumoutzis, K.; Petanidou, T. Short- and Mid-Term Spatiotemporal Diversity Patterns of Post-Fire Insect-Pollinated Plant Communities in the Mediterranean. *Land* **2023**, *12*, 2109. [CrossRef]
127. Papavassiliou, S.; Arianoutsou, M. Natural Post-Fire Regeneration of Leguminosae in a Pinus Halepensis Forest of Attica. In Proceedings of the European School of Climatology and Natural Hazards Course, Porto Carras, Halkidiki, Greece, 27 May–4 June 1997; Balabanis, P., Eftichidis, G., Fantechi, R., Eds.; European Commission: Brussels, Belgium, 1997; pp. 411–419.
128. Poirazidis, K.; Chaideftou, E.; Martinis, A.; Bontzorlos, V.; Galani, P.; Kalivas, D. Temporal Shifts in Floristic and Avian Diversity in Mediterranean Pine Forest Ecosystems under Different Fire Pressure: The Island of Zakynthos as a Case Study. *Ann. For. Res.* **2017**, *61*, 19–36. [CrossRef]
129. Radea, C.; Arianoutsou, M. Soil Arthropod Communities and Population Dynamics Following Wildfires in Pine Forests of the Mediterranean Basin: A Review. *Isr. J. Ecol. Evol.* **2012**, *58*, 137–149.
130. Radea, C.; Kazanis, D.; Arianoutsou, M. Effects of Fire History Upon Soil Macroarthropod Communities in Pinus Halepensis Stands in Attica, Greece. *Isr. J. Ecol. Evol.* **2010**, *56*, 165–179. [CrossRef]
131. Spanos, I.; Raftoyannis, Y.; Goudelis, G.; Xanthopoulou, E.; Samara, T.; Tsiontsis, A. Effects of Postfire Logging on Soil and Vegetation Recovery in a Pinus Halepensis Mill. Forest of Greece. *Plant Soil.* **2005**, *278*, 171–179. [CrossRef]
132. Verroios, G.; Georgiadis, T. Post-Fire Vegetation Succession. The Case of Aleppo Pine (Pinus Halepensis Miller) Forest of Northern Achaia, (Greece). *Fresenius Environ. Bull.* **2002**, *11*, 186–193.
133. El Khayati, M.; Chergui, B.; Barranco, P.; Fahd, S.; Ruiz, J.L.; Taheri, A.; Santos, X. Assessing the Response of Different Soil Arthropod Communities to Fire: A Case Study from Northwestern Africa. *Fire* **2023**, *6*, 206. [CrossRef]
134. Agbeshie, A.A.; Abugre, S.; Atta-Darkwa, T.; Awuah, R. A Review of the Effects of Forest Fire on Soil Properties. *J. For. Res.* **2022**, *33*, 1419–1441. [CrossRef]
135. Huerta, S.; Marcos, E.; Fernández-García, V.; Calvo, L. Short-Term Effects of Burn Severity on Ecosystem Multifunctionality in the Northwest Iberian Peninsula. *Sci. Total Environ.* **2022**, *844*, 157193. [CrossRef]
136. Mayer, M.; Baltensweiler, A.; James, J.; Rigling, A.; Hagedorn, F. A Global Synthesis and Conceptualization of the Magnitude and Duration of Soil Carbon Losses in Response to Forest Disturbances. *Glob. Ecol. Biogeogr.* **2024**, *33*, 141–150. [CrossRef] [PubMed]
137. Xofis, P.; Buckley, P.G.; Kefalas, G.; Chalaris, M.; Mitchley, J. Mid-Term Effects of Fire on Soil Properties of North-East Mediterranean Ecosystems. *Fire* **2023**, *6*, 337. [CrossRef]
138. Rodrigues, M.; De La Riva, J.; Domingo, D.; Lamelas, T.; Ibarra, P.; Hoffrén, R.; García-Martín, A. An Empirical Assessment of the Potential of Post-Fire Recovery of Tree-Forest Communities in Mediterranean Environments. *For. Ecol. Manag.* **2024**, *552*, 121587. [CrossRef]
139. Turner, M.G.; Braziunas, K.H.; Hansen, W.D.; Harvey, B.J. Short-Interval Severe Fire Erodes the Resilience of Subalpine Lodgepole Pine Forests. *Proc. Natl. Acad. Sci. USA* **2019**, *116*, 11319–11328. [CrossRef]
140. Eckdahl, J.A.; Kristensen, J.A.; Metcalfe, D.B. Restricted Plant Diversity Limits Carbon Recapture after Wildfire in Warming Boreal Forests. *Commun. Earth Environ.* **2024**, *5*, 186. [CrossRef]
141. Moghli, A.; Santana, V.M.; Baeza, M.J.; Pastor, E.; Soliveres, S. Fire Recurrence and Time Since Last Fire Interact to Determine the Supply of Multiple Ecosystem Services by Mediterranean Forests. *Ecosystems* **2022**, *25*, 1358–1370. [CrossRef]
142. Peris-Llopis, M.; Vastaranta, M.; Saarinen, N.; González-Olabarria, J.R.; García-Gonzalo, J.; Mola-Yudego, B. Post-Fire Vegetation Dynamics and Location as Main Drivers of Fire Recurrence in Mediterranean Forests. *For. Ecol. Manag.* **2024**, *568*, 122126. [CrossRef]
143. Wilkinson, D.M. The Disturbing History of Intermediate Disturbance. *Oikos* **1999**, *84*, 145. [CrossRef]
144. Athanasopoulou, E.; Rieger, D.; Walter, C.; Vogel, H.; Karali, A.; Hatzaki, M.; Gerasopoulos, E.; Vogel, B.; Giannakopoulos, C.; Gratsea, M.; et al. Fire Risk, Atmospheric Chemistry and Radiative Forcing Assessment of Wildfires in Eastern Mediterranean. *Atmos. Environ.* **2014**, *95*, 113–125. [CrossRef]
145. Kaskaoutis, D.G.; Petrinoli, K.; Grivas, G.; Kalkavouras, P.; Tsagkaraki, M.; Tavernaraki, K.; Papoutsidaki, K.; Stavroulas, I.; Paraskevopoulou, D.; Bougiatioti, A.; et al. Impact of Peri-Urban Forest Fires on Air Quality and Aerosol Optical and Chemical Properties: The Case of the August 2021 Wildfires in Athens, Greece. *Sci. Total Environ.* **2024**, *907*, 168028. [CrossRef] [PubMed]
146. Lazaridis, M.; Latos, M.; Aleksandropoulou, V.; Hov, Ø.; Papayannis, A.; Tørseth, K. Contribution of Forest Fire Emissions to Atmospheric Pollution in Greece. *Air Qual. Atmos. Health* **2008**, *1*, 143–158. [CrossRef]
147. Martinho, V.J.P.D. Estimating Relationships between Forest Fires and Greenhouse Gas Emissions: Circular and Cumulative Effects or Unidirectional Causality? *Environ. Monit. Assess.* **2019**, *191*, 581. [CrossRef] [PubMed]
148. Poupkou, A.; Markakis, K.; Liora, N.; Giannaros, T.M.; Zanis, P.; Im, U.; Daskalakis, N.; Myriokefalitakis, S.; Kaiser, J.W.; Melas, D.; et al. A Modeling Study of the Impact of the 2007 Greek Forest Fires on the Gaseous Pollutant Levels in the Eastern Mediterranean. *Atmos. Res.* **2014**, *149*, 1–17. [CrossRef]
149. Sazeides, C.I.; Christopoulou, A.; Fyllas, N.M. Coupling Photosynthetic Measurements with Biometric Data to Estimate Gross Primary Productivity (GPP) in Mediterranean Pine Forests of Different Post-Fire Age. *Forests* **2021**, *12*, 1256. [CrossRef]

150. Molina, J.R.; Herrera, M.A.; Rodríguez, Y.; Silva, F. Wildfire-Induced Reduction in the Carbon Storage of Mediterranean Ecosystems: An Application to Brush and Forest Fires Impacts Assessment. *Environ. Impact Assess. Rev.* **2019**, *76*, 88–97. [CrossRef]
151. Kotsiras, K.; Kokkoris, I.P.; Strid, A.; Dimopoulos, P. Integrating Plant Diversity Data into Mapping and Assessment of Ecosystem and Their Services (MAES) Implementation in Greece: Woodland and Forest Pilot. *Forests* **2020**, *11*, 956. [CrossRef]
152. Moreira, F.; Allsopp, N.; Esler, K.J.; Wardell-Johnson, G.; Ancillotto, L.; Arianoutsou, M.; Clary, J.; Brotons, L.; Clavero, M.; Dimitrakopoulos, P.G.; et al. Priority Questions for Biodiversity Conservation in the Mediterranean Biome: Heterogeneous Perspectives across Continents and Stakeholders. *Conserv. Sci. Prac.* **2019**, *1*, e118. [CrossRef]
153. Depietri, Y.; Orenstein, D.E. Fire-Regulating Services and Disservices With an Application to the Haifa-Carmel Region in Israel. *Front. Environ. Sci.* **2019**, *7*, 107. [CrossRef]
154. Ameztegui, A.; Coll, L.; Cáceres, M.D.; Morán-Ordóñez, A. Disturbance Impacts on Mediterranean Forests across Climate and Management Scenarios. *J. Environ. Manag.* **2024**, *371*, 123193. [CrossRef] [PubMed]
155. Picard, N.; Garavaglia, V. Mediterranean Forests and the United Nations Sustainable Development Goals. In *Pines and Their Mixed Forest Ecosystems in the Mediterranean Basin*; Ne'eman, G., Osem, Y., Eds.; Managing Forest Ecosystems; Springer International Publishing: Cham, Switzerland, 2021; Volume 38, pp. 603–616; ISBN 978-3-030-63624-1.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.